The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence*

Matteo Iacoviello[†] Jonathan Tong[‡]

June 13, 2026

Abstract

We introduce an improved measure of geopolitical risk, constructed by applying artificial intelligence to newspaper content. Our approach replaces keyword matching with semantic understanding: instead of searching for specific word combinations, we use OpenAI's GPT-4o-mini model to read newspaper articles and assess their geopolitical risk intensity. The daily AI-GPR index scores about 4.6 million articles from the New York Times, Washington Post, and Chicago Tribune from 1960 through 2025. The approach reduces false positives from articles mentioning war or terrorism in non-geopolitical contexts while capturing relevant articles that lack exact or common dictionary terms. The AI-GPR index also assigns gradations of risk intensity rather than simple yes-or-no classifications, providing more nuanced measurement even at high frequencies. We demonstrate the potential of our approach with several applications. The AI-GPR index produces a precisely estimated negative effect of geopolitical risk on stock returns, and decomposing it into anticipated threats and realized acts reveals that markets price the two differently. A second LLM classification layer yields a historical time series of geopolitical risk-driven oil supply disruptions by region, which we use to trace the macroeconomic effects of geopolitical oil shocks. Finally, the layered architecture decomposes geopolitical risk by country and by directed country pair, and we show that the resulting bilateral index predicts the contraction in trade between countries in tension.

JEL Classification: D80, E66, F51, G15.

Keywords: Geopolitical Risk, Text Analysis, Large Language Models, Measurement Error, Content Classification.

---

*We thank Flora Haberkorn, the Proquest TDM Studio team, and seminar participants at the Federal Reserve Board, the Financial Risks Forum in Paris, and University of Naples Partenope for helpful comments. The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of anyone else associated with the Federal Reserve System. The data presented in this paper are available at `https://www.matteoiacoviello.com/ai_gpr.html`.

[†]Federal Reserve Board of Governors and CEPR. Email: matteo.iacoviello@frb.gov

[‡]University of Wisconsin. Email: jtong45@wisc.edu

1 Introduction

Geopolitical risk—encompassing wars, terrorism, and tensions between nations—is an important driver of macroeconomic fluctuations. Central to the empirical understanding of its effects is the ability to quantify latent political tension from sparse and noisy data. Caldara and Iacoviello (2022) established a framework for this measurement with the Geopolitical Risk (GPR) Index, using dictionary-based keyword counts to track global instability through newspaper archives.[1] Their index has since become a standard benchmark in the literature, providing a transparent and computationally efficient proxy for the risks that shape trade patterns, policy uncertainty, investor sentiment, and asset prices.

While keyword-based methods have proven highly effective at capturing the broad contours of global risk and other economic phenomena, they are subject to the inherent constraints of dictionary matching. These methods may struggle to distinguish between articles where a geopolitical event is the primary subject from those where keywords appear in a peripheral or historical context. Furthermore, fixed dictionaries may miss evolving terminology or fail to capture the nuanced intensity of diplomatic language. This paper builds directly upon the conceptual architecture of Caldara and Iacoviello (2022), but proposes a methodological refinement: the transition from keyword matching to semantic understanding.

We introduce the AI-GPR Index, constructed by applying a Large Language Model (LLM)—specifically OpenAI's GPT-4o-mini—to a corpus of approximately 4.6 million articles from the *New York Times*, *Washington Post*, and *Chicago Tribune* between 1960 and 2025. Our approach maintains the definitions established in the prior literature but utilizes the LLM's contextual awareness to score articles on a continuous scale (0.0 to 1.0). This allows for a more granular identification of risk, distinguishing articles where a geopolitical event is central from articles in which geopolitical risk is only peripherally mentioned. The resulting index complements the original GPR by offering a smoother and more accurate time series that retains the transparency of the original framework through auditable, prompt-based classification criteria.

We demonstrate the value of the AI-GPR index through a number of empirical applications. First, we revisit the relationship between geopolitical risk and financial markets. We find that the AI-GPR index provides a more precisely estimated negative effect of geopolitical innovations on stock market returns, particularly when isolating the persistent component of risk. A further classification layer decomposes the index into anticipated threats and realized acts, and we show that the two are priced differently: markets re-price gradually to the persistent buildup of threats, but respond to realized acts only insofar as they are unanticipated. Second, we employ a layered classification architecture to identify articles specifically related to geopolitical oil supply disruptions.

[1] The use of dictionary-based methods itself builds on the foundational work of Baker, Bloom, and Davis (2016) and Hassan et al. (2019), among others.

This yields a novel, 65-year time series of GPR-driven oil shocks disaggregated by geographic region. We validate the Oil GPR index using a proxy structural VAR, in the spirit of Verduzco-Bustos and Zanetti (2026), but replacing their original GPR instrument with a novel proxy constructed from oil price surprises on days when the Oil GPR index spikes; the identified shock generates a persistent rise in oil prices and a significant contraction in economic activity.

Third, we leverage the LLM's ability to parse complex linguistic structures to identify the country-level actors involved in each adverse geopolitical event and their specific roles—as initiators, respondents, or spillover parties. We use this information to construct country-level indices of geopolitical risk, tracing which nations drive global risk over time and extending the country sub-indices of Caldara and Iacoviello (2022) to the full set of nations mentioned in the news.

Fourth, we provide a bilateral decomposition of geopolitical risks by constructing country-level bilateral indices, representing an important advance relative to prior methods. While keyword-based approaches are limited to identifying the co-occurrence of country names within a text, our methodology explicitly identifies directed actions between states. This allows us to construct high-frequency, directed country-pair indices of geopolitical stress that capture the source and target of geopolitical tensions, providing a clearer view of bilateral relationships across countries that was previously inaccessible at this scale. By mapping these directed links, we can identify the specific country pairs that drive aggregate global risk. We validate the bilateral index by showing that it predicts bilateral trade flows: in a standard gravity equation, a one-standard-deviation increase in bilateral GPR is associated with roughly a 0.04 standard-deviation decline in bilateral trade the following year, a result that is stable across specifications.

We make all the measures presented in this paper available to the research community in a companion webpage. The scope of these new measures has itself been shaped by that community: the high-frequency country-level decompositions, the oil-market measures of geopolitical risk, and the bilateral stress indicators introduced in this paper each respond to requests and suggestions that have accumulated over the years from researchers, policymakers and market participants who have relied on and popularized the original index of Caldara and Iacoviello.

Our work contributes to a rapidly emerging literature applying LLMs to economic measurement (e.g., Ito, Sato, and Ota, 2025 and Clayton, Coppola, Maggiori, and Schreger, 2025). We demonstrate that LLM-based approaches do not replace, but rather augment, the dictionary-based methods that have defined the field. As computational costs continue to decline, we believe this high-resolution approach to text analysis will become a standard template for measuring many complex political and economic phenomena that are otherwise hard to quantify. Our focus on geopolitical risk also connects our work to the growing literature on geoeconomics (see Mohr and Trebesch, 2025 and Clayton, Maggiori, and Schreger, 2026) and provides an improved set of data to make sense of geopolitical risk both within and across country borders.

The paper proceeds as follows. Section 2 describes the data and the LLM-based methodology.

Section 3 presents the AI-GPR index and compares it with the keyword-based benchmark. Section 4 introduces three further dimensions of measurement—threats and acts, oil, and country and bilateral risk. Section 5 presents our empirical applications, and Section 6 concludes.

2 Data and Methodology

2.1 Data Sources

Our construction of the AI-GPR index sources newspaper data from the New York Times, Washington Post, and Chicago Tribune, all of which provide comprehensive international coverage and maintain consistent editorial standards over time. The articles we score span from 1960 to 2025 and are sampled daily.[2]

We adopt a two-stage process that first filters for articles likely to discuss geopolitical events and subsequently passes them through an LLM for scoring. The advantages of this method are twofold. First, the keyword screen efficiently discards the many articles—given the broad coverage of newspapers—that carry no discussion of geopolitical risk at all. Second, for the articles that pass the screen, the LLM classifies geopolitical risk content and intensity more accurately than query-based alternatives.

We use the following query on the ProQuest TDM Studio database to filter for articles likely to discuss geopolitical risk:

```
SEARCH QUERY:

PUBDATE(>=19600101) AND PUBDATE(<=20251231) AND LA(English) AND
((airstrike* OR alliance OR annex* OR attack* OR blockade* OR bomb* OR
cease-fire OR combat OR confrontation OR conflict* OR coup OR crisis OR
diplomat* OR diplomacy OR embargo OR enem* OR hostage* OR hostil* OR
instability OR invade* OR invasion* OR militia OR military OR missile* OR
nuclear OR refugee* OR riot* OR sanction* OR sovereign* OR terror* OR
treaty OR troop* OR truce OR unrest OR violence OR war OR weapon*)) AND
(publication("New York Times") OR publication("Washington Post") OR
publication("Chicago Tribune")) AND DTYPE(article OR commentary OR
editorial OR feature OR 'front page article' OR 'front page/cover story' OR
news OR report OR review)
```

[2] The headline corpus counts, summary statistics, and computational-cost figures reported in this paper refer to the 1960–2025 period covered by the search query below. For the time-series figures and the financial-market regressions of Section 5.1, we extend the index through March 2026 by applying the identical sampling, screening, and scoring pipeline to articles published through that date, changing only the upper bound on the publication date. The companion webpage hosts the regularly updated series.

The query focuses on articles containing at least one keyword related to geopolitical risk. We further restrict our attention to articles categorized as news content rather than advertisements or non-editorial material. Our keywords span three categories of geopolitical risk: direct conflict and military action (war, invasion*, attack*, terror*, conflict*, ...), political crisis and instability (crisis, instability, unrest, coup...), and strategic responses and diplomacy (sanction*, embargo, blockade*...). Our search extracts 4.6 million articles from the universe of articles published across the three publications over the sample period. Of these 4.6 million articles, 1.2 million (26 percent) receive a positive GPR score from the LLM. Section 2.5 verifies that the keyword filter has a negligible false-negative rate.

For each selected article, we extract metadata (publication date, headline) and its textual content from the ProQuest database. To manage computational costs, we truncate article text to the first 2,000 characters. Due to the inverted pyramid structure of journalism where the most important information is presented first, the truncation captures the essential geopolitical content for the vast majority of articles. As we show in Section 2.5, our baseline scores based on truncated text are nearly identical to those obtained using the full article text.

2.2 LLM Classification: Definition, Prompt, Implementation

We use Caldara and Iacoviello (2022)'s definition of geopolitical risk, which encompasses wars (initiation of new military conflicts between states or organized groups), escalation of existing wars (intensification of ongoing military conflicts), major terrorist attacks (violent acts intended to intimidate civilian populations or governments), and tensions among states and political actors (diplomatic disputes, threats, military buildups, and other developments that affect the peaceful course of international relations). This definition provides clear boundaries for what constitutes geopolitical risk while encompassing the full range of relevant events. Importantly, it focuses on current risks and actual events, not historical retrospectives, fictional depictions, or metaphorical uses of geopolitical terminology.

We use the following system prompt to instruct the LLM to evaluate articles:

```
CLASSIFICATION PROMPT:
You will be given a news article.  Classify the article's assessment of
geopolitical risk based only on what the article states or strongly implies.
Geopolitical risk is defined as the threat, realization, and escalation of adverse
events associated with:  wars (initiation of new conflicts); escalation of
existing wars; major terrorist attacks; tensions among states and political actors
that affect the peaceful course of international relations.
Assign a geopolitical risk score from 0.0 to 1.0 based on the following scale:
0.0--0.2:  No mention of geopolitical risks.
```

```
0.2--0.4: Mentions of minor tensions, diplomatic disputes, or isolated incidents
with limited escalation risk.
0.4--0.6: Discussion of significant tensions, ongoing regional conflicts, or
moderate escalation risk.
0.6--0.8: Substantial discussion of major war initiation/escalation risks, active
terrorism threats, or high likelihood of significant escalation.
0.8--1.0: Extensive coverage of imminent or new war, major war escalation, severe
terrorism threats, or critical threat to international stability.
Movies, books, anniversaries of old events, and obituaries should receive a score
of 0.0 unless they explicitly mention current risks.
Note: Purely domestic political events (elections, protests, internal policy
debates) should score 0.0 unless they have implications for international tensions
or cross-border conflicts.
```

As the prompt illustrates, we use zero-shot prompting: the model receives explicit criteria and a scoring scale but no labeled examples. One-shot or few-shot prompting would require embedding one or more example articles in every request; at the scale of approximately 4.6 million articles, even a single example would roughly double input token costs. The zero-shot approach keeps prompt length short and cost proportional to the number of articles, while the detailed instruction set substitutes for examples by giving the model precise, auditable criteria. The scale from 0.0 to 1.0 allows the model to rate articles with varying intensities of geopolitical risk rather than binary classification.

We implement the model with temperature set to zero, which minimizes stochastic variation in responses and ensures consistent scoring across multiple runs. We use structured output to return model output in JSON format, making extraction and processing of scores more efficient.[3]

For each article, we construct a request consisting of the system prompt and the article's textual content (the headline and first 2,000 characters of the article). The API returns a JSON object containing the geopolitical risk score. We validate that all scores fall within the specified 0.0–1.0 range.

2.3 Index Construction

We construct the AI-GPR index by aggregating individual article scores at daily frequency. For each date t, we compute:

$$\text{AI-GPR}_t = \frac{1}{\bar{S}} \times \frac{1}{A_t} \sum_{i=1}^{N_t} S_{it} \tag{1}$$

where S_{it} represents the geopolitical risk score for article i at time t, N_t is the total number of

[3] Structured output is particularly valuable for classification tasks with multiple output dimensions, such as the second-stage classifiers introduced in Section 4.

scored articles published on date t, A_t is the total number of articles published on date t across all three newspapers (not just those matching geopolitical keywords), and $\bar{S}$ is a normalization constant chosen such that the index has a mean of 100 over the 1985–2019 period. Dividing by A_t accounts for variation in newspaper output over time and is adopted from Caldara and Iacoviello (2022) for consistency.[4]

Having the AI-GPR index aggregated daily allows it to capture high-frequency fluctuations in geopolitical risk. Unlike monthly or quarterly indices, the daily frequency preserves information about short-lived risk spikes and allows researchers to better identify the specific timing of geopolitical events and their economic effects. This precise temporal resolution is particularly valuable for event studies and examining the immediate market reactions to geopolitical developments.

2.4 Comparison with Keyword-Based Approaches

Our LLM-based approach differs from traditional keyword-based classification methods in several fundamental ways. Caldara and Iacoviello (2022) classify articles by searching for specific combinations of words from predefined categories. For example, an article is classified as relevant to "war threats" if it contains a geopolitical term ("war," "conflict," "hostilities") within a specified proximity of a risk term ("threat," "danger," "crisis"). To reduce false positives, the original GPR methodology employs exclusion words like "movie," "anniversary," and "obituary" that disqualify an article from classification even if it contains the requisite geopolitical keywords.

This exclusion strategy addresses some common sources of false positives but remains imperfect. Articles about war films may not use the word "movie" and still discuss fictional content. Historical retrospectives may avoid words like "anniversary" while still focusing on past events rather than current events. Moreover, the binary nature of their exclusion methodology (article is either included or excluded entirely) cannot handle articles that legitimately discuss both current geopolitical risks and historical context.

Our approach handles these cases more flexibly by leveraging the LLM's semantic understanding. Rather than applying binary exclusion rules, the model evaluates whether geopolitical content is central to the article and whether it pertains to current rather than historical events. An article that mentions a war film in passing while primarily analyzing current military tensions would receive an appropriate non-zero score, whereas an article primarily reviewing a war movie would correctly receive a zero score.

Beyond the false positive and false negative concerns, keyword-based approaches face additional limitations. First, keyword matching is context-insensitive: it cannot distinguish whether geopolit-

[4] The denominator A_t counts articles containing at least seven common words ("the," "be," "to," "of," "and," "at," and "in"). This choice filters out non-editorial content such as photos with a brief caption, classified advertisements, table-of-contents entries, stock quotes, sports results, and brief notices, and serves as a normalization for trends in newspaper article volume over the years. The total number of articles containing all these words is about 8.5 million. Were we to count every object classified as an article, the total would be almost twice as large.

ical content is central or peripheral to an article, so an article titled "The Price of War in Ukraine" receives the same binary classification as one mentioning "the war on inflation" in a subordinate clause. Keyword-based matching weighs every occurrence of a geopolitical term equally, so it tends to overvalue the geopolitical content of articles that happen to contain such terms. Second, the approach is terminology-dependent, requiring exhaustive keyword lists; articles discussing "armed confrontation" or "military standoff" may be missed if these exact phrases are not in the dictionary or if the proximity requirements are not satisfied. In practice, it is neither realistic nor feasible to create a query that accounts for all possible geopolitical terms used in newspapers, so some articles will always be missed with keyword-based approaches. Third, binary classification is not able to differentiate the intensity or centrality of geopolitical content. Our LLM-based approach addresses these three limitations through semantic understanding. The model evaluates whether geopolitical content is central to the article, uses context to distinguish current risks from historical references, recognizes semantically relevant content regardless of specific terminology, and provides graduated scores reflecting severity. Rather than applying rigid exclusion rules, the model exercises contextual judgment about relevance, which we believe resolves the biggest flaws of the original approach.

The tradeoff is reduced transparency. While keyword lists and exclusion rules are fully auditable, LLM-based classification relies on complex neural networks and is effectively a black box. To introduce as much predictability and human oversight as possible into the LLM's classifications, we make deliberate choices in how we prompt and call the model. Our prompt-based instruction set provides explicit criteria, and we set the temperature of the model to 0 to minimize output randomness. The model's decisions can be examined by reviewing its classifications on validation samples, and from our manual auditing of sample articles, the model's scoring is consistent with our own judgment.

2.5 Measurement Error

Three potential sources of measurement error affect the AI-GPR index: model stochasticity, text truncation, and attenuation from the two-stage filtering process. We separately validate the index against human judgment in Section 2.6.

Model stochasticity. Even at temperature zero, LLMs are not deterministic and may produce slightly different outputs across calls. To quantify this, we run GPT-4o-mini three times on the same subset of 1,050 articles. The correlation between the first two runs is 0.99, with 98 percent of articles receiving identical scores and a mean absolute difference of 0.004. The correlation between the first and the third run is also 0.99, with 98.6 percent of articles receiving identical scores and a mean absolute difference of 0.003. This near-perfect reproducibility confirms that classification noise from model stochasticity is negligible. To further bound this concern, we re-ran the model at temperatures of 0.5 and 1.0 on the same articles. Within-run agreement remains above 91 percent even at temperature 1.0, and cross-temperature agreement between temperature 0.0 and temperature 1.0 is 93.2 percent with a mean absolute difference of 0.014. Mean scores are virtually indistinguishable across temperature settings, confirming that the choice of temperature has negligible impact on classification outcomes.[5] We further bound this concern through the multi-model comparison reported in Section 3.4: pairwise correlations across four OpenAI models and an open-weight Llama model—spanning two generations, different model sizes, and different model families—range from 0.85 to 0.94, implying that classification is driven by article content rather than idiosyncratic model behavior.

Text length. Our baseline classifier truncates article text at 2,000 characters to reduce processing time and cost. To verify that this truncation does not materially affect classification, we re-score the same 1,050 articles using the complete article text. The correlation between truncated and full-text scores is 0.93, with 86.7 percent of articles receiving identical scores. Full-text scores are modestly higher on average (mean 0.153 versus 0.129), consistent with longer articles containing additional geopolitical content beyond the opening paragraphs, but the cross-article ranking is nearly identical. We conclude that truncation introduces minimal classification error.

First-stage attenuation. The keyword pre-filter could introduce measurement error by missing geopolitically relevant articles (false negatives) or by including irrelevant ones (false positives). We assess the false-negative rate by applying the LLM classifier to a random sample of articles that did *not* match any of our search keywords. Only 0.9 percent received a positive geopolitical risk score, and those that did had uniformly low scores, confirming that the keyword filter captures the vast majority of geopolitically relevant content with a negligible false-negative rate. This compares favorably with the 2.6 percent false-negative rate reported by Caldara and Iacoviello (2022),[6] whose keyword filter permanently excludes articles without a second-stage review. False positives from our keyword filter are handled naturally by the LLM scoring step: articles that match keywords but do not discuss geopolitical risk receive a score of zero and do not contribute to the index.

2.6 Validation: Human Audit

To assess whether the model systematically over- or under-scores articles relative to a human reader, we independently scored a random sample of 1,050 articles by hand, applying the same 0-to-1 rubric given to the LLM.[7] The human and machine scores line up closely: their correlation is 0.88, they agree on whether an article contains any geopolitical risk in 92 percent of cases, and they fall within

[5] Correlation and agreement across temperatures remain robust when compared to the human-annotated sample. The correlation between human annotations and model scores remains very high and similar across the temperature 0.5 and temperature 1.0 runs, as does the agreement rate.

[6] See Appendix B.6 in Caldara and Iacoviello (2022).

[7] Full details of the audit—the cross-tabulation of human and machine scores, a taxonomy of the disagreements, and illustrative cases—are reported in Appendix A.2.

one step of the scoring scale (0.2) of each other for 98 percent of articles. Because the AI-GPR is a continuous score rather than a binary label, classification-error rates are only a coarse summary of agreement; the correlation and the near-exact concordance above are the more natural metrics. They can nonetheless be compared with keyword-based selection: dichotomizing the scores at any positive value, about 12 percent of the articles the model flags are ones the human scores as zero—a false-discovery rate below the 21 percent that Caldara and Iacoviello (2022) report for their keyword search.[8] The mean machine score slightly exceeds the mean human score, a mild upward bias, and the agreement is stable over time, with decade-by-decade correlations between 0.84 and 0.91 and no discernible trend.

Because the two scores diverge on only about one article in four, almost always by a single step of the scale, the disagreements are few enough to read individually, and doing so is informative about how to interpret the machine index. The model's mild upward bias comes mainly from two recurring patterns: it tends to read cross-border economic and financial disputes—tariffs, currency crises, frayed trade "ties"—and episodes of purely domestic unrest as geopolitical risk, even though the rubric reserves the concept for war, terrorism, and military tension. Its errors of timing are roughly offsetting: it sometimes scores stale, retrospective coverage as though the events were current, and sometimes misses the fresh aftermath of an attack, so the two biases largely cancel in the aggregate. At the top of the scale the model is conservative, rarely using the highest gradations, so it modestly understates the intensity of the most extreme episodes. A further set of disagreements arise where the human credited context that an article only implies, and so reflect ambiguity in the coding rubric as much as model error. On balance, with a correlation of 0.88 and near-universal agreement to within one step of the scale, the audit supports treating the AI-GPR scores as a reliable proxy for careful human coding, with the caveat that the index runs slightly hot at the bottom of the scale and slightly cold at the top.

Carlson and Dell (2025) propose a statistical correction for indices built from text classifiers. Their key insight is that keyword- or model-based classifiers produce biased proxies for the true concept of interest, and that this bias can be removed by combining classifier predictions with a human-labeled validation sample that covers the full span of the data. Applied to the GPR index of Caldara and Iacoviello, they find that the keyword-based series underestimates geopolitical risk once the correction is applied. A practical challenge with this approach is that the validation sample must be large enough and representative enough to pin down the bias over time, which can require extensive hand-annotation. Our LLM approach addresses the underestimation problem directly at the classification stage—by replacing keyword matching with semantic understanding—and produces a similarly elevated series without requiring a post-hoc statistical correction. The

[8] This rate is the share of *flagged* articles—those receiving a positive score—that the human scores as zero, the quantity comparable to the figure Caldara and Iacoviello report, which is likewise defined over the articles their keyword search selects. Measured instead as a share of all audited articles, the false-positive rate is 3.6 percent.

stable human-AI agreement across decades documented above (correlations 0.84–0.91) suggests that residual bias in the AI-GPR is small and does not vary systematically over time, limiting the scope for the kind of time-varying underestimation that the Carlson and Dell correction is designed to address.

2.7 Computational Considerations

A practical concern for LLM-based approaches is computational cost. At current pricing, processing one article through GPT-4o-mini costs approximately $0.0001 for our truncated text length. For our sample of approximately 4.6 million articles from 1960–2025, the total computational cost is approximately $450. These costs will continue to decline as API pricing falls, which should enable larger and more widespread usage of LLMs for sentiment analysis.

We also find that the processing time is manageable. With parallelization across multiple API calls, we can process the full corpus in approximately 250 hours of computation time. These practical considerations suggest that LLM-based text classification is increasingly feasible for large-scale economic research applications, including the construction of long historical time series.[9]

3 The AI-GPR Index

3.1 Summary Statistics

Table 1 presents summary statistics for the AI-GPR index alongside a keyword-based GPR index over the 1960–2025 period. The keyword-based index applies the Caldara and Iacoviello (2022) methodology—which classifies articles using proximity searches of geopolitical and risk-related terms—to the same three newspapers used for the AI-GPR (New York Times, Washington Post, and Chicago Tribune). Caldara and Iacoviello (2022) publish two versions of the index—a historical index based on three newspapers that extends back to 1900, and a recent index based on ten newspapers that begins in 1985—and our comparison is with the three-newspaper version, which we compute on the same three newspapers as the AI-GPR back to 1960 to ensure a like-for-like comparison. The full search string is reported in Appendix A.3. Both indices are normalized so that the mean equals 100 over 1985–2019.

Several features of the AI-GPR index stand out. It has a lower standard deviation than the original GPR (48.5 versus 60.6) and a thinner right tail (a 99th percentile of 267 versus 300), reflecting its greater smoothness. Its first percentile sits well above zero (42 versus 18), and the index records no zero days, consistent with the fact that news articles always carry some low-level geopolitical discussion even on quiet days. The higher 90-day autocorrelation of the AI-GPR (0.73 vs. 0.62) indicates greater persistence: geopolitical conditions evolve smoothly rather than spiking

[9] Using other models, such as the GPT-5 mini, may be more costly or require longer processing times.

and reverting.[10] The correlation between the two indices is 0.69, indicating that they capture similar dynamics while exhibiting meaningful differences.[11]

Figure 1 displays the AI-GPR index as a 30-day moving average over the full sample, with labels marking the major geopolitical episodes of the past six decades.

3.2 Comparison with the Original GPR Index

Figure 2 plots the AI-GPR alongside the original GPR index at monthly frequency. The two series co-move closely overall but exhibit notable divergences.

In the 1960s–1970s, the AI-GPR index is elevated relative to the original, consistent with the AI model's ability to detect geopolitical risk in articles that use language conventions of the era that may not match the keyword dictionary. The Cold War-era tensions, the Vietnam War, and the Yom Kippur War all register prominently. In the 1990s, the AI-GPR index is higher during the Bosnian War (1992–1995), suggesting that the LLM captures the sustained geopolitical significance of this conflict more effectively than keyword counting. In the 2000s, both indices spike sharply after the September 11 attacks and during the Iraq War. The AI-GPR exhibits a somewhat faster decline after the initial Iraq War spike, consistent with the LLM's ability to distinguish between articles primarily covering active conflict versus articles mentioning the Iraq war in other contexts. From 2010 onward, the indices track each other closely, though the AI-GPR is somewhat more elevated in recent years. This may reflect the LLM's ability to capture emerging geopolitical risks (e.g., cyberwarfare, economic coercion) that do not always trigger traditional keyword matches.

As conjectured by Carlson and Dell (2025), the AI-GPR index finds that the share of articles discussing geopolitical risks is higher than in the original GPR tabulations. As shown in Table 1, the AI-GPR classifies 15 percent of all articles as mentioning geopolitical risks, compared to just 3.3 percent under the original, keyword-based approach.[12] Nevertheless, the original GPR retains considerable validity as a measure of geopolitical risk: the two series share a correlation of 0.69 and track each other closely around major events, suggesting that while the keyword approach may understate the share of geopolitically relevant newspaper articles, it still captures the same peaks

[10] We measure persistence as the autocorrelation of each index's 90-day moving average: specifically, the correlation between the smoothed series on day t and on day $t-90$. This statistic captures slow-moving variation in geopolitical conditions rather than day-to-day volatility, and is consistent with the definition reported in Table 1.

[11] A comparison of classification rates with Caldara and Iacoviello (2022) can be reconciled as follows. Caldara and Iacoviello (2022) classify 3.3 percent of articles as GPR-relevant, but report a 21 percent Type-I error rate and a 2.6 percent false-negative rate among excluded articles. Correcting for both, their implied true GPR rate is approximately 5.1 percent ($= 3.3 \times 0.79 + 96.7 \times 0.026$). In our approach, 26 percent of sampled articles (15 percent of all articles) receive a positive score, but two-thirds of these receive scores of 0.2 or 0.4, reflecting low-intensity geopolitical risk content that keyword matching systematically misses. Only one-third—approximately 9 percent of sampled articles, or 5 percent of all articles—score above 0.5. This high-intensity subset is directly comparable to Caldara and Iacoviello's adjusted rate of 5.1 percent, suggesting both approaches identify a similar core of high-salience geopolitical articles. Small differences in these rates reflect the fact that Caldara and Iacoviello's validation statistics refer to their ten-newspaper index over 1985–2019, while ours covers three newspapers from 1960–2025.

[12] Among articles sent to the LLM, 26.2 percent receive a positive score.

and troughs that define the phenomenon.

3.3 Performance on Historical Events

Figure 3 compares the AI-GPR and original GPR around four major geopolitical events: the Cuban Missile Crisis (1962), the Kuwait Invasion (1990), the September 11 Attacks (2001), and Russia's invasion of Ukraine (2022). The daily (non-smoothed) indices reveal important differences.

For more recent episodes—September 11 and the Russia-Ukraine invasion—the AI-GPR index shows less day-to-day volatility while still capturing the spike in risk. One possible explanation is that modern journalistic language is more readily interpretable by the LLM: the AI model can parse contemporary writing style more accurately than a keyword-based approach that relies on the exact appearance of specific terms. For earlier episodes such as the Cuban Missile Crisis, the AI-GPR also captures the risk elevation but with a smoother profile, likely because the semantic approach is less sensitive to day-to-day variation in keyword frequency.

3.4 Model Comparison

To assess whether our choice of LLM affects the resulting index, we classify a random sample of 1,050 articles using four OpenAI models—GPT-4o-mini (our baseline), GPT-4o, GPT-5-mini, and GPT-5—and an open-weight model, Llama 3.1 8B Instruct. All models receive the same prompt and are run at temperature zero.[13] The pairwise correlations between GPT-4o-mini scores and each alternative model are 0.90 (GPT-4o), 0.87 (GPT-5-mini), 0.86 (GPT-5), and 0.91 (Llama).[14] The share of articles classified as containing geopolitical risk (score $>$ 0) is remarkably similar across models, ranging from 26 to 33 percent. These results indicate strong agreement across models spanning two generations, different model sizes, and both proprietary and open-weight model families, suggesting that the AI-GPR index is not sensitive to the specific model used. Given that GPT-4o-mini is substantially cheaper and produces highly correlated scores, we adopt it as our baseline for the full corpus.

4 The Anatomy of Geopolitical Risk

We further leverage the semantic understanding of LLMs to measure three additional dimensions of geopolitical risk: threats and acts, oil, and country-level and bilateral risk. The threats-and-acts measure identifies whether geopolitical risk is anticipated or realized. The oil measure captures the intensity of oil-supply disruptions driven by geopolitical events. The country and bilateral measures track which countries are involved in geopolitical events and the directed relationships

[13] By default, GPT-5-mini and GPT-5 have temperature set to 1.0, and this parameter cannot be changed. Instead, we set reasoning effort to high.

[14] Within the GPT-5 family, the correlation between GPT-5-mini and GPT-5 is 0.94.

between them. Each is produced by applying an additional classification layer to the underlying articles.

4.1 Threats and Acts

A key innovation of Caldara and Iacoviello (2022) was to separate geopolitical risk into *threats*—adverse events that are anticipated, feared, or warned of but have not yet materialized—and *acts*—events that have actually occurred or are underway. The distinction is economically meaningful, because anticipated and realized risk need not move the economy in the same way, as our analysis of stock returns in Section 5.1 makes clear. We build an analogous decomposition of the AI-GPR index with a second LLM classification layer that reads each article and judges the temporal nature of the risk it describes.

We apply the classifier to every article that receives a positive AI-GPR score, passing the same text supplied to the baseline classifier to GPT-4o-mini at temperature zero.[15] The model places each article in one of three categories: a *threat*, when it centers on risks, warnings, tensions, or potential escalations that have not yet come to pass; an *act*, when it reports an adverse event that has occurred or is occurring, such as an attack, invasion, or coup; or *both*, when it substantively discusses a realized event together with the risks of further escalation that surround it. Because articles in the *both* category speak to each dimension, we count them toward both subindexes, so that the threat series aggregates the articles labeled *threat* or *both* and the act series aggregates those labeled *act* or *both*.[16] Each subindex is then built exactly like the headline AI-GPR index: we sum the geopolitical-risk scores of its articles, divide by total newspaper volume, and rescale the series to average 100 over 1985–2019.

Figure 4 plots the two subindexes at monthly frequency, and they trace distinct geopolitical rhythms. Acts spike sharply around realized conflicts—the 1991 Gulf War, the September 11 attacks, the 2003 Iraq War, and Russia's 2022 invasion of Ukraine—while threats run higher during the early Cold War and other spells of elevated but unrealized tension, when coverage is dominated by warnings and brinkmanship rather than open conflict. As Section 5.1 shows, this distinction is more than descriptive: threats and acts are priced quite differently in equity markets.

4.2 Oil

We construct a second subindex about oil supply disruptions driven by geopolitical events using a two-step classification process similar to Section 4.1. In the first step, we restrict attention to articles receiving a GPR score above 0.5 and, to economize on classification costs, retain those

[15] As in the baseline scoring step, the model reads the first 2,000 characters of each article. The full prompt is reproduced in Appendix A.4.

[16] About 15 percent of classified articles fall in the *both* category, with roughly 48 percent labeled threats and 37 percent acts.

mentioning at least one oil- or energy-related keyword; articles without such keywords are recorded as carrying no oil-supply risk. In the second step, we apply a second LLM prompt to the retained articles.[17] This prompt asks GPT-4o-mini to determine whether the article discusses an oil or energy supply disruption caused by a geopolitical event and, if so, which geographic region(s) are affected from a predefined list: Middle East, Russia, USA, Venezuela, North Africa, West Africa, Central Asia, North Sea, Canada, Mexico, Latin America, Southeast Asia, and China.[18]

For each date t, we construct the oil disruption ratio:

$$\text{Oil GPR}_t = \frac{\sum_{i \in D_t} S_{it}}{A_t} \tag{2}$$

where D_t is the set of articles on date t classified as discussing a geopolitical oil disruption, S_{it} is the article's GPR score (from the first classification layer), and A_t is the total number of newspaper articles published on date t. The numerator weights each disruption article by its GPR intensity, so that an article extensively covering an oil crisis (score near 1.0) contributes more than one with a passing mention (score near 0.5). The denominator normalizes by total publishing volume, analogously to the construction of the AI-GPR index. Regional oil disruption indices are constructed similarly, restricting D_t to articles classified as affecting a particular region.

Figure 5 plots the Oil GPR index (30-day moving average, monthly) alongside real WTI crude oil prices over 1960–2025. The index closely tracks the major geopolitical oil supply events: the 1973 Arab oil embargo, the 1979 Iranian Revolution, the 1990–91 Gulf War, and the Russia-Ukraine conflict and associated sanctions from 2022 onward. The co-movement between the Oil GPR index and oil prices confirms that the LLM-based classification is capturing the geopolitical episodes most associated with supply disruptions. A regional decomposition of the Oil GPR index by geography—showing the shifting dominance of the Middle East through the 1990s, Russia after 2014, and Venezuela during its political crises—is provided in Figure A.1 of the appendix.

Table 3 reports summary statistics for the oil disruption classification. Of all articles with a GPR score above 0.5, about 13 percent contain oil-related keywords and are sent to the second classification layer. Of these, roughly three-quarters are classified as discussing a geopolitical oil supply disruption. The Middle East accounts for nearly two-thirds of all disruption mentions, consistent with its dominant role in geopolitical oil risk throughout the sample period. Russia is the second most frequently mentioned region at 14 percent, with its share rising sharply after 2014.

4.3 Country and Bilateral

Our final set of measures focuses on country-level and bilateral geopolitical risk.

[17] To improve classification accuracy, the second-stage classifiers pass the first 3,000 characters of each article to GPT-4o-mini, compared to 2,000 characters for the base GPR scoring step.

[18] The full classification prompt is located in Appendix A.4.

The country subindex measures the intensity of geopolitical events associated with a particular country. To construct the subindex, we parse each article through a classification layer to identify which countries are involved in the geopolitical event and their corresponding actor roles. We aggregate the AI-GPR scores for each country across all roles and scale to the overall index.[19] Figure 7 presents monthly series for the United States, Russia, China, and the United Kingdom—the four countries most consistently featured in the geopolitical risk literature. The companion webpage includes time series of geopolitical risk for 193 countries.

The United States is the most persistently prominent country, reflecting its role as the global hegemon engaged in or affected by conflicts worldwide throughout the sample. Russia registers two distinct elevated periods: the Cold War era through the late 1980s, and the post-2014 period following the annexation of Crimea and the full-scale invasion of Ukraine in 2022. China's GPR contribution rises steadily from the late 1990s onward, consistent with its growing global presence and the intensification of geopolitical competition in the Indo-Pacific. Country-level decompositions of this kind extend the country sub-indices of Caldara and Iacoviello (2022), who constructed keyword-based series for selected nations. The LLM approach enables a more precise construction of sub-indices across all countries mentioned in the news without requiring pre-specified keyword lists.

We build on the country subindex to create the bilateral index. This index tracks the level of geopolitical risk between initiator-respondent country pairs. We construct bilateral indices for the top 1,200 country pairs as the sum of AI-GPR sentiment scores in articles where the first country is classified as initiator and the second as respondent, so that the USA→Russia and Russia→USA series are distinct.[20]

This subindex represents an innovation that would not have been possible using past keyword approaches, which could identify that Russia and the United States both appeared in an article but could not distinguish whether Russia was acting on the United States, whether the United States was responding to Russian pressure, or whether both countries merely appear in adjacent paragraphs about unrelated topics. The LLM classifier extracts the directed relational structure of each event, identifying which country initiates the geopolitical action and which is the target.

Figure 8 presents the 12-month moving averages for the nine top-ranked directed pairs by average GPR intensity over 1960–2025. The United States appears as initiator in seven of the nine pairs, reflecting both its central role in postwar geopolitics and the US-centric newspaper sources underlying the index. The pairs span the Cold War confrontation with the Soviet Union (USA→Russia, Russia→USA, USA→Cuba), the major US military interventions of the postwar

[19] The full country classifier prompt is in Appendix A.4.

[20] Each directed-pair index is scaled analogously to the country-level indices: the raw sentiment sum for the pair is multiplied by the ratio of the overall AI-GPR index to the total coded sentiment in that month. Because an article in which one country acts against multiple respondents contributes its full sentiment score to each initiator–respondent pair (and vice versa), the bilateral indices do not in general sum to the aggregate index.

era (USA→Vietnam, USA→Iraq, USA→Afghanistan), ongoing great-power and regional tensions (USA→China, USA→Iran), and Russia's 2022 invasion of Ukraine (Russia→Ukraine).

The same classification layer also labels each article by the type of geopolitical event it describes—among them military conflict, diplomatic tension, terrorism, civil war, nuclear threat, coup or regime change, and sanctions.[21] Although we do not pursue these categories as a separate application, they line up with familiar historical patterns: military conflict tracks the major wars of the postwar era, peaking around the Korean and Vietnam Wars, the Gulf War, and Russia's 2022 invasion of Ukraine; terrorism rises sharply after the September 11 attacks; and sanctions grow steadily more prominent after 2014 as economic coercion becomes a favored geopolitical instrument.

5 Geopolitical Risk in Markets, Oil, and Trade

We present a series of applications identifying causal effects of the three dimensions of geopolitical risk we have studied thus far. First, we quantify the effects of geopolitical threats and acts on the stock market using the threats and acts subindexes. We then explore the effects of oil shocks on the macroeconomy by constructing a novel instrument from the Oil GPR index. Lastly, we study how shocks in bilateral relations affect international trade using our bilateral GPR index.

5.1 Threats and Acts and Financial Markets

A large literature studies whether geopolitical events depress stock markets. We revisit this question using the AI-GPR index and daily data on U.S. stock returns from 1960 to March 2026.

We merge the AI-GPR index with daily excess market returns.[22] Since stock markets are closed on weekends and holidays while the AI-GPR is computed for every calendar day, we construct a trading-day GPR measure as follows. If the market is closed for k consecutive calendar days before trading day t, we assign to day t the average of the AI-GPR over those $k+1$ days (including day t itself), so that geopolitical information accumulating over non-trading days is correctly attributed to the next trading day. We then work with the first difference of this series, standardized by its full-sample standard deviation. All GPR variables in the regressions below are standardized in this way. We estimate the regression:

$$r_t^e = \alpha + \beta\,\Delta\mathrm{GPR}_t + \varepsilon_t \tag{3}$$

where r_t^e is the excess market return (in percent). Table 2 reports the results estimated on data

[21] The event-type classification prompt is listed in Appendix A.4.

[22] Through November 28, 2025, we use data from the Fama-French daily factors dataset. See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Beginning December 1, 2025, when Fama-French daily factors are not yet available, we extend the return series using the daily excess return on the S&P 500 index from Haver Analytics, computed as the daily simple return on the index minus the daily-compounding equivalent of the three-month Treasury bill rate.

collapsed to weekly frequency, where stock market returns are cumulated weekly and the GPR change is computed week over week. The relationship is negative and economically meaningful: a one-standard-deviation increase in the AI-GPR is associated with a decline of about fourteen basis points in the weekly market return.

The OLS estimate in equation (3) conflates predictable and unpredictable movements in GPR. To disentangle these, we decompose geopolitical risk into anticipated and unanticipated components. We estimate an AR(4) model for the standardized GPR change:

$$\Delta\mathrm{GPR}_t = \phi_0 + \sum_{j=1}^{4} \phi_j \, \Delta\mathrm{GPR}_{t-j} + u_t \tag{4}$$

and decompose $\Delta\mathrm{GPR}_t$ into the persistent (predictable) component $\widehat{\Delta\mathrm{GPR}}_t$ and the shock (unpredictable) component $\hat{u}_t$. We then estimate:[23]

$$r_t^e = \alpha + \beta_1 \, \widehat{\Delta\mathrm{GPR}}_t + \beta_2 \, \hat{u}_t + \varepsilon_t \tag{5}$$

We estimate both regressions—the level specification in equation (3) and the anticipated/unanticipated decomposition in equation (5)—twice: first using the aggregate AI-GPR index, and then replacing it with the threat and act subindexes entered jointly. The two pairs of regressions map onto the four columns of Table 2—the aggregate index in columns (1) and (2), and its threat-and-act refinement in columns (3) and (4).

For the aggregate index, the predictable, persistent component of geopolitical risk depresses returns far more than the unexpected component—its effect is more than twice as large—and both are statistically significant. This finding has a clear economic interpretation. While a sudden geopolitical event like the September 11 attacks generates an immediate negative return, the "long tail" of that risk, the fact that geopolitical risk remains high in the weeks that follow, is what sustains an elevated equity risk premium. Investors do not merely react to headlines; they aggressively re-price assets as geopolitical uncertainty becomes embedded in the macroeconomic outlook.

The gap between the muted aggregate OLS estimate and the much larger effect of the persistent component suggests that a simple linear regression understates the role of geopolitical risk, attenuated by the high-frequency noise of weekly shocks, which are large in magnitude but transient in their effect on asset prices.[24] After decomposing the risk, the market is roughly twice as sensitive to the persistent component of geopolitical risk than the raw OLS estimate suggests.

[23] This approach follows the empirical tradition of Barro (1979), who decomposes macroeconomic variables into anticipated and unanticipated components to test whether agents respond differently to predictable trends versus stochastic innovations.

[24] This result echoes the findings in Gonçalves et al. (2025), who examine the pricing of geopolitical risk using stock market returns. They use Caldara and Iacoviello's decomposition of geopolitical risk into forward-looking threats (GPT) and realized acts (GPA), and find that only the threat component is consistently priced across assets and forecasts equity premia, whereas realized acts have a much weaker and less stable link to risk premia.

Separating the index into its threat and act components sharpens this picture. In levels, it is geopolitical *threats* whose effect on returns is precisely estimated, while realized *acts*—though comparable in magnitude—are too noisy to register as statistically significant. The decomposition into anticipated and unanticipated movements reveals why the two are priced so differently. Threats bite through their predictable part: as a threat persists and works its way into the outlook, markets re-price gradually, so it is the slow-moving, anticipated component of threat risk that weighs on returns. Acts work the other way around. Because events that can be foreseen are already embedded in prices, only the *unanticipated* occurrence of an act moves markets—and it is indeed the surprise in acts, not their predictable part, that carries the effect. This same asymmetry explains why acts seem to do nothing in the simple regression: pooling a meaningful surprise effect together with a negligible anticipated one averages the act response toward zero, the very dilution that mutes the aggregate estimate. The threat-and-act results thus reinforce the central lesson of the aggregate decomposition: what matters for asset prices is not the level of geopolitical risk but whether it is anticipated or unexpected—a distinction that a single regression cannot capture.

The improved day-to-day precision of the AI-GPR index is central to these results. When we repeat the same analysis using the keyword-based GPR index, the results are qualitatively similar but economically and statistically weaker: the contemporaneous relationship with returns is roughly a third smaller and only marginally significant, and in the decomposition the persistent component barely clears conventional significance while the shock component loses it altogether. These differences illustrate how measurement imprecision in the keyword-based index attenuates and weakens the estimated effects. Keyword-based methods misclassify articles at higher rates, so the resulting index contains more variation that is not geopolitically relevant, biasing coefficients toward zero in a standard errors-in-variables fashion. The AI-GPR's semantic scoring reduces this day-to-day noise, yielding a sharper signal that strengthens the estimated relationship between geopolitical risk and stock returns.

5.2 Oil Market Shocks and Macro Effects

The Oil GPR index provides a natural instrument for isolating oil price movements driven by geopolitical supply disruptions, as distinct from demand-driven fluctuations or weather-related supply shocks. We borrow this idea from Verduzco-Bustos and Zanetti (2026), who use innovations in the Caldara and Iacoviello (2022) GPR index to identify geopolitically driven oil shocks; we replace their measure with the oil-specific Oil GPR index, which leads to better inference by targeting exactly the subset of geopolitical news related to energy supply.[25] We estimate a four-variable proxy

[25] As an additional step, we construct the monthly instrument by summing daily WTI log-price changes on days when the Oil GPR index spikes above two standard deviations, shifted back one day to account for the newspaper reporting lag. This ensures the identified shock reflects oil price movements triggered by geopolitical news rather than pre-existing price trends.

structural VAR (Mertens and Ravn, 2013; Stock and Watson, 2012) over the period 1975–2024 at monthly frequency. The system includes the nominal WTI oil price, global oil production, U.S. monthly GDP (the Brave-Butters-Kelley index, Brave et al. 2019), and the Oil GPR index. The VAR is estimated with 12 lags, and the external instrument—monthly WTI oil price surprises on Oil GPR event days—identifies the structural geopolitical oil price shock.

Figure 6 presents the impulse responses to a geopolitical oil supply shock, normalized to generate a 33 percent increase in oil prices on impact. Oil production declines on impact, consistent with a supply disruption. U.S. GDP falls by roughly 1 percent over one to two years, reflecting the contractionary effect of higher energy costs on economic activity. The Oil GPR index itself spikes on impact and then gradually dissipates as tensions ease. These results confirm that the Oil GPR index isolates economically meaningful supply-side disturbances, and that geopolitically driven oil shocks have quantitatively significant macroeconomic effects.

5.3 Bilateral Shocks and International Trade

The bilateral index enables a direct test of whether geopolitical risk depresses bilateral trade. We merge our bilateral GPR series for up to 1,200 country pairs with the CEPII Gravity database (Conte et al., 2023), which provides annual bilateral trade flows from BACI—covering up to 200 countries—and standard gravity controls over 1960–2019. We include all pairs for which we construct bilateral GPR (up to 1,200) and for which matching trade data are available in BACI.[26] Both bilateral trade and bilateral GPR are standardized within country pair to remove pair-specific means and scales, and GPR enters with a one-year lag. Table 4 reports the estimates. Across three specifications—year fixed effects alone (column 1), year fixed effects plus log PPP-adjusted GDP of both countries (column 2), and origin-by-year plus destination-by-year fixed effects (column 3)—a one-standard-deviation increase in bilateral GPR is associated with approximately a 0.04 standard-deviation decline in bilateral trade in the following year. Figure 9 displays the partial correlation after removing year effects and GDP controls.

6 Conclusion

We have introduced the AI-GPR Index, an improved measure of geopolitical risk that uses large language models to evaluate newspaper content. Our approach maintains the conceptual framework established by Caldara and Iacoviello (2022) while improving measurement precision through semantic understanding rather than keyword matching.

The methodology offers several practical advantages. LLM-based classification reduces false positives from topically irrelevant keyword matches and false negatives from semantically relevant

[26] Trade flows are deflated to constant 2012 dollars using the US GDP deflator.

articles using alternative terminology. The approach provides graduated risk scores rather than binary classification, allowing the index to reflect intensity as well as presence of geopolitical content. Unlike rigid exclusion rules that disqualify entire articles based on specific words, our method exercises contextual judgment about whether geopolitical content is central and current. The method extends straightforwardly to non-English publications and other text sources.

We illustrated several applications of the index. First, the AI-GPR produces a more precisely estimated negative effect of geopolitical risk on stock returns and reveals that markets are especially sensitive to the persistent component of GPR innovations; decomposing the index into threats and acts shows that this persistent effect runs through anticipated threats, while the surprise component of realized acts also weighs on prices. Second, by combining the AI-GPR with a second classification layer, we constructed a novel historical time series of geopolitical oil supply disruptions by region, revealing the shifting geography of energy-related geopolitical risk over six decades and the macroeconomic effects of geopolitically driven oil shocks. Third, by extracting the geopolitical actors in each event and their roles, we decomposed risk to the level of individual countries, extending the analogous sub-indices of Caldara and Iacoviello (2022) using richer LLM-based scores. Fourth, the bilateral dimension is genuinely new, identifying which countries are acting on whom rather than merely co-occurring in coverage, and we show that bilateral GPR predicts bilateral trade in a standard gravity framework, with a one-standard-deviation increase in geopolitical risk associated with roughly a 0.04 standard-deviation decline in trade the following year.

Our main contribution is demonstrating that LLM-based approaches can be applied at scale to construct consistent historical time series spanning multiple decades. As computational costs continue to decline, LLM-based approaches become increasingly practical for large-scale text analysis in economics. The approach is complementary to existing keyword-based methods, offering researchers and policymakers an additional tool for tracking geopolitical risk dynamics across time and space.

Future research could extend this methodology to construct firm-level or sector-level measures of geopolitical risk exposure, analyze how geopolitical risks transmit across countries and markets using the bilateral and country-level indices, or examine the relationship between geopolitical risks and various economic outcomes. The flexibility of prompt-based classification extends naturally to a wide range of economic and social phenomena beyond geopolitical risk, and we hope this framework offers a useful template for such work.

References

Baker, S. R., N. Bloom, and S. J. Davis (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics 131*(4), 1593–1636.

Barro, R. J. (1979). Unanticipated money growth and unemployment in the united states: Reply. *The American Economic Review 69*(5), 1004–1009.

Brave, S. A., R. A. Butters, and D. Kelley (2019). A new "big data" index of U.S. economic activity. *Economic Perspectives 43*(1). Federal Reserve Bank of Chicago.

Caldara, D. and M. Iacoviello (2022). Measuring geopolitical risk. *American Economic Review 112*(4), 1194–1225.

Carlson, J. and M. Dell (2025). A unifying framework for robust and efficient inference with unstructured data. arXiv:2505.00282.

Clayton, C., A. Coppola, M. Maggiori, and J. Schreger (2025, June). Geoeconomic pressure. Working Paper.

Clayton, C., M. Maggiori, and J. Schreger (2026). A framework for geoeconomics. *Econometrica 94*(1), 105–136.

Conte, M., P. Cotterlaz, and T. Mayer (2023). The CEPII gravity database. Working paper, CEPII.

Gonçalves, A. S., A. Melone, and A. Ricciardi (2025). The pricing of geopolitical tensions over a century. *Fisher College of Business Working Paper*. WP 2025-03-010.

Hassan, T. A., S. Hollander, L. van Lent, and A. Tahoun (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics 134*(4), 2135–2202.

Ito, A., M. Sato, and R. Ota (2025). A novel content-based approach to measuring monetary policy uncertainty using fine-tuned llms. *Finance Research Letters 75*, 106832.

Mertens, K. and M. O. Ravn (2013). The dynamic effects of personal and corporate income tax changes in the United States. *American Economic Review 103*(4), 1212–1247.

Mohr, C. and C. Trebesch (2025). Geoeconomics. *Annual Review of Economics 17*.

Stock, J. H. and M. W. Watson (2012). Disentangling the channels of the 2007–2009 recession. *Brookings Papers on Economic Activity 2012*(1), 81–135.

Verduzco-Bustos, G. and F. Zanetti (2026). The effects of geopolitical oil price shocks. Working paper.

Table 1: AI-GPR Index: Summary Statistics

Statistic	**AI-GPR**	**GPR (Original)**
Total newspaper articles	8,514,822	8,514,822
Articles sampled	4,571,496	—
Articles scored positive (> 0)	1,166,083	262,096
Standard Deviation	48.52	60.55
Skewness	1.64	1.97
1st Percentile	41.97	18.14
25th Percentile	79.61	68.22
Median	105.46	99.15
75th Percentile	138.33	139.28
99th Percentile	266.95	300.16
90-day Autocorrelation	0.73	0.62
Correlation with Original GPR	0.69	1.00
Positive Articles Share (%)	15.00	3.28
Positive Among Sampled (%)	26.16	3.28

Notes: Summary statistics computed over the full 1960–2025 daily sample. The AI-GPR index is computed as the sum of LLM-assigned geopolitical risk scores for all articles on each date, normalized by total newspaper article count. The "GPR (Original)" column is the three-newspaper version of the Caldara and Iacoviello (2022) GPR index—their historical index, which extends back to 1900—computed here on the same three newspapers as the AI-GPR (*New York Times*, *Washington Post*, and *Chicago Tribune*) over 1960–2025; the comparison is with this three-newspaper version rather than their ten-newspaper index, which begins in 1985. Both indices are normalized to a mean of 100 over 1985–2019. The 90-day autocorrelation is the correlation between the 90-day moving average on date t and on date $t-90$. Positive Articles Share (%) is the average daily share of all newspaper articles classified as containing geopolitical risk: for the AI-GPR, articles receiving a score > 0; for the original GPR, articles matching the keyword proximity search. Both are expressed as a share of total newspaper articles published. Positive Among Sampled (%) is the average daily share of positive articles among those in the queried sample: for the AI-GPR, the share of articles sent to the LLM (i.e., matching the broad keyword filter) that receive a score > 0; for the original GPR, this coincides with Positive Articles Share (%) since there is no separate sampling stage. The first three rows report totals over the full sample: total newspaper articles (the normalization denominator), articles sampled (sent to the LLM for scoring; the original GPR has no separate sampling stage), and articles scored positive (AI-GPR score > 0, or a keyword match for the original GPR).

Table 2: Weekly Stock Returns and Geopolitical Risk

	Aggregate GPR		Threats and Acts	
	(1) OLS	(2) Decomposition	(3) OLS	(4) Decomposition
ΔGPR_t	-0.136^{**} (0.057)			
$\Delta\text{GPR}_{t,\,threats}$			-0.081^{**} (0.037)	
$\Delta\text{GPR}_{t,\,acts}$			-0.084 (0.061)	
Persistent $\widehat{\Delta\text{GPR}}_t$		-0.271^{**} (0.134)		
Persistent $\widehat{\Delta\text{GPR}}_{t,\,threats}$				-0.220^{**} (0.103)
Persistent $\widehat{\Delta\text{GPR}}_{t,\,acts}$				-0.063 (0.131)
Shock $\hat{u}_t$		-0.119^{*} (0.062)		
Shock $\hat{u}_{t,\,threats}$				-0.052 (0.044)
Shock $\hat{u}_{t,\,acts}$				-0.090^{**} (0.042)
Constant	0.137^{***} (0.037)	0.138^{***} (0.037)	0.136^{***} (0.037)	0.138^{***} (0.037)
Observations	3,452	3,448	3,452	3,448
R^2	0.004	0.004	0.004	0.004
Sample	1960–2026	1960–2026	1960–2026	1960–2026

Notes: Weekly regressions of excess market returns (r_t^e, in percent) on standardized changes in the AI-GPR index. Columns (1)–(2) use the aggregate AI-GPR index; columns (3)–(4) use its threats and acts subindexes. Weeks run Monday through Sunday; weekly stock returns are the sum of daily excess returns within the week, and the weekly GPR variable is the change in the within-week average of the trading-day GPR measure, standardized over the full sample. Columns (1) and (3) report OLS regressions of r_t^e on the GPR change(s), as in equation (3). Columns (2) and (4) decompose each GPR change into a persistent component (fitted values from an AR(4) model) and a shock component (residuals), as in equation (5). Excess market returns are from the Fama-French daily factors dataset through November 28, 2025, extended with Haver SP500 excess returns (SP500 simple return minus the daily-compounded 3-month T-bill rate) from December 1, 2025 onward. OLS columns report heteroskedasticity-robust standard errors in parentheses; decomposition columns report pairs-bootstrap standard errors (1,000 replications) to account for the generated regressors in the two-step procedure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: GPR-Driven Oil Disruption Classification: Summary Statistics

Statistic	**Value**
Classification rates (% of articles with GPR score > 0.5*):*	
Containing oil/energy keywords	12.6%
Classified as oil disruption	9.3%
Of keyword articles classified as disruption	73.8%
Disruptions by region (% of disruption articles):	
Middle East	62.9%
Russia	14.1%
North Africa	10.8%
USA	5.7%
Southeast Asia	5.8%
West Africa	5.7%
Latin America	5.2%
China	3.8%
Venezuela	3.0%
Other	3.8%

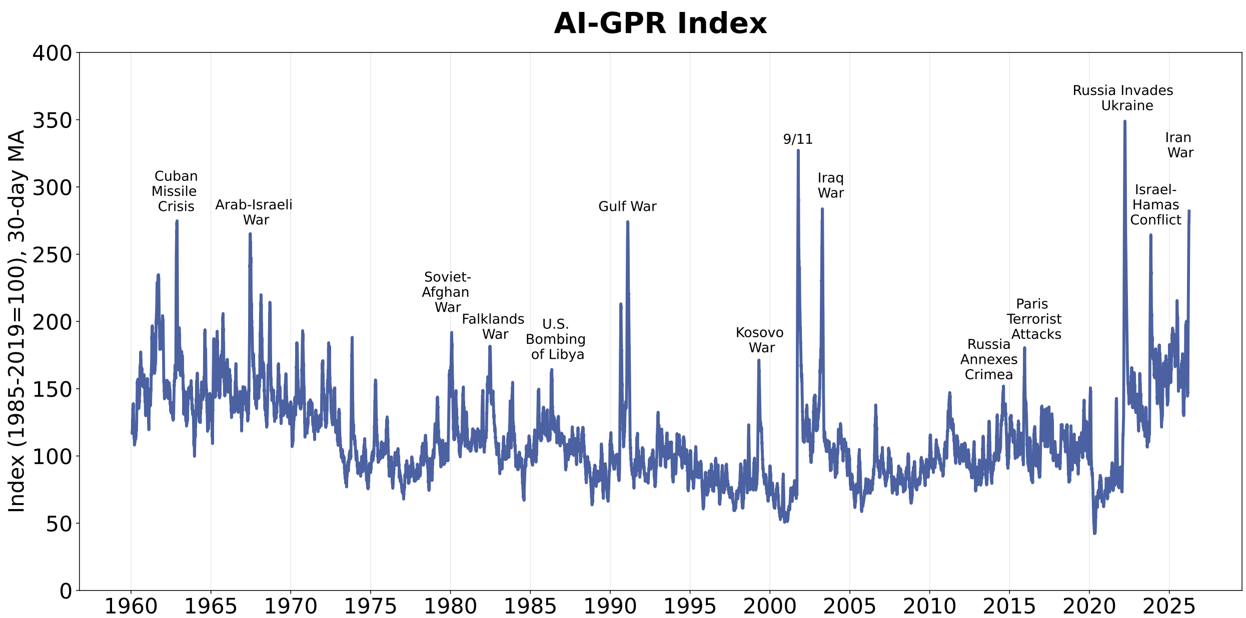
Notes: Summary statistics for the two-stage oil disruption classification. Articles with a GPR score above 0.5 that contain at least one oil- or energy-related keyword are classified by a second LLM prompt to determine whether the article discusses an oil or energy supply disruption driven by a geopolitical event and, if so, which region(s) are affected; articles without such keywords are recorded as non-disruptions. Regional percentages sum to more than 100% because articles may mention multiple regions. "Other" includes Central Asia, Mexico, North Sea, and Canada. Sample: New York Times, Washington Post, and Chicago Tribune, 1960–2025.

Table 4: Bilateral Geopolitical Risk and Trade Flows

	(1) Year FE	(2) Year FE + GDP	(3) Country×Year FE
Bilateral GPR_{t-1} (std.)	-0.043^{***}	-0.040^{***}	-0.039^{***}
	(0.012)	(0.013)	(0.015)
ln GDP PPP, origin		0.016^{***}	
		(0.003)	
ln GDP PPP, destination		0.021^{***}	
		(0.003)	
Observations	16,708	12,648	12,648

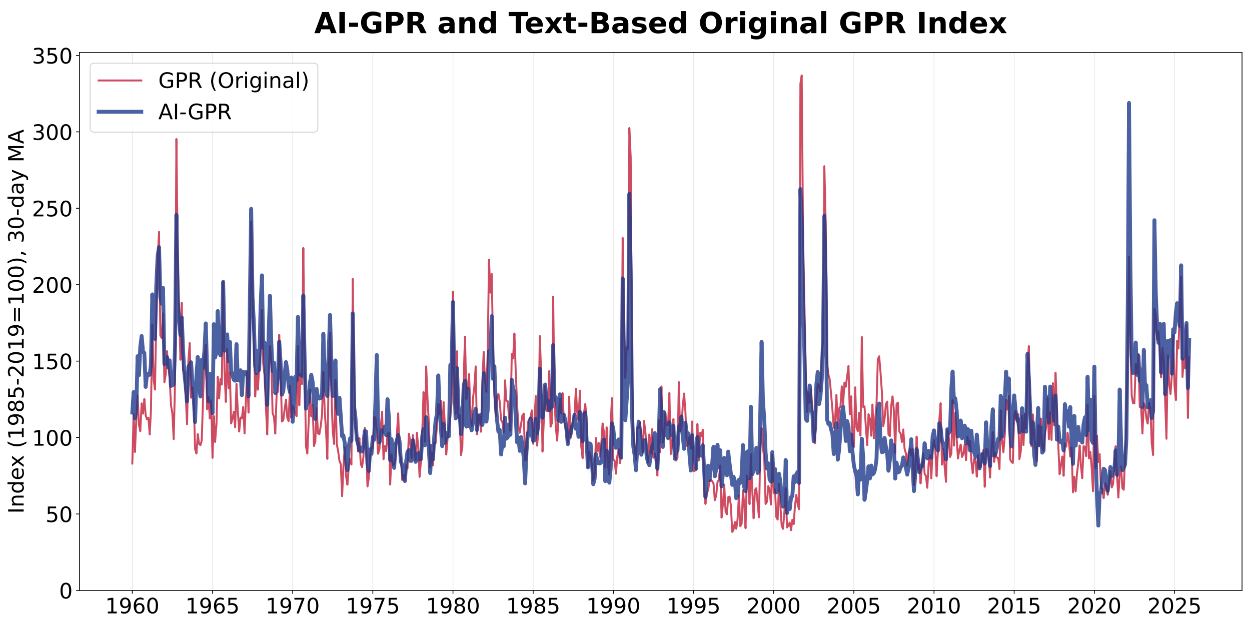
Notes: Dependent variable is bilateral trade (BACI, deflated to 2012 USD), standardized within country pair. Bilateral GPR is standardized within pair and enters with a one-year lag. Column 1 includes year fixed effects. Column 2 adds log PPP-adjusted GDP (PWT) for origin and destination. Column 3 uses origin-by-year and destination-by-year fixed effects, which absorb GDP. Standard errors clustered by country pair. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: bilateral GPR from this paper; trade and gravity controls from Conte et al. (2023).

Figure 1: The AI-GPR Index (30-Day Moving Average)



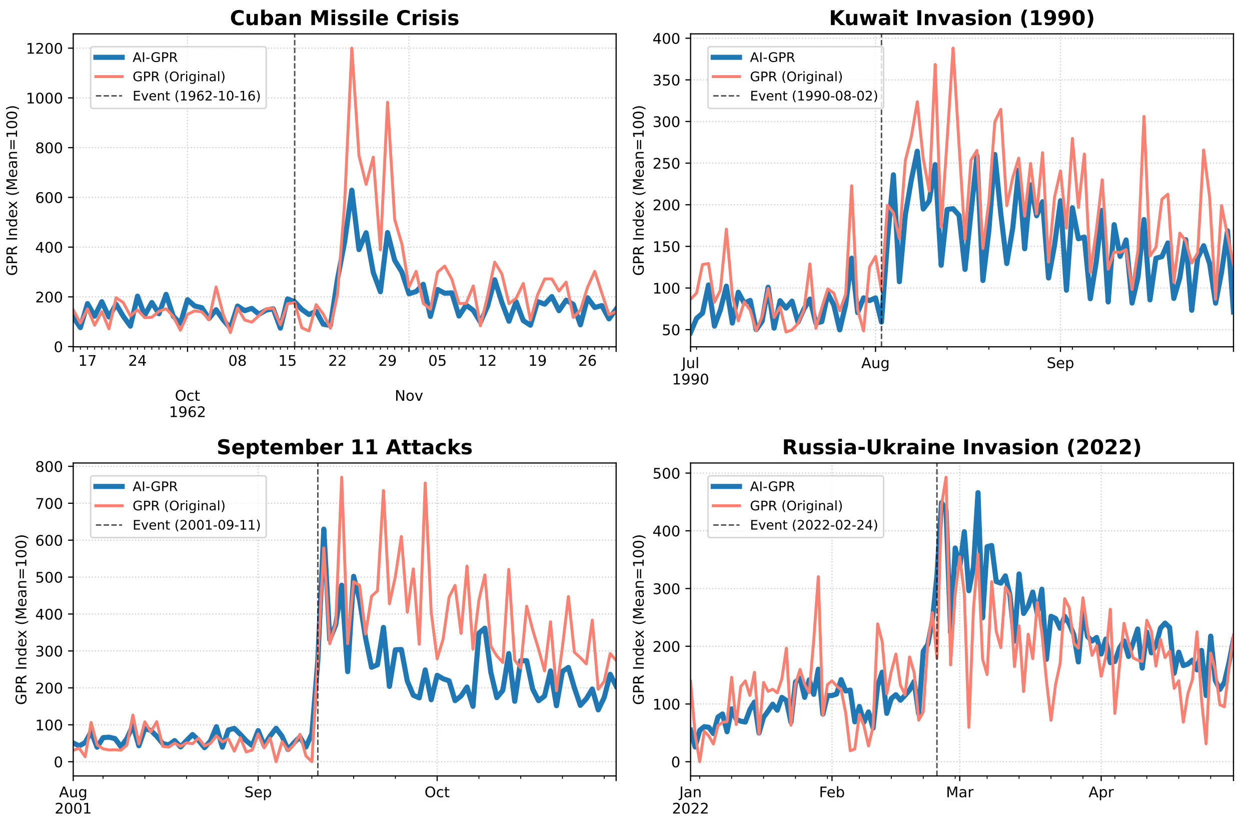
Notes: The AI-GPR index plotted as a 30-day moving average, from January 1960 through March 31, 2026. The index sums LLM-assigned geopolitical risk scores (0–1) across all articles each day, normalized by total newspaper article count, and is scaled to a mean of 100 over 1985–2019. Regularly updated data are available on the companion webpage. Source: authors' calculations using ProQuest TDM Studio data.

Figure 2: AI-GPR and Original GPR Index (Monthly)



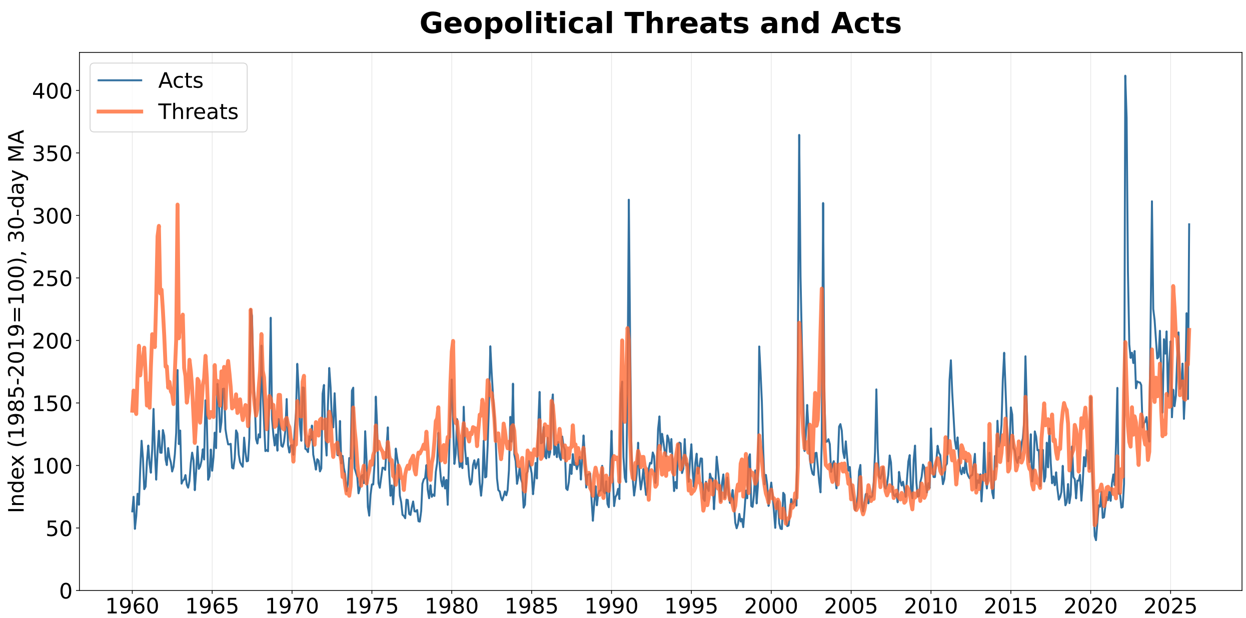
Notes: The AI-GPR index (blue, thick) and the original GPR index (red, thin), both plotted at monthly frequency, 1960–2025. The original GPR here is the three-newspaper version of the Caldara and Iacoviello (2022) index—their historical index, which extends back to 1900—computed on the same three newspapers as the AI-GPR; the comparison is with this version rather than their ten-newspaper index, which begins in 1985. Both indices are normalized to a mean of 100 over 1985–2019. The AI-GPR index sums LLM-assigned geopolitical risk scores across all articles each day, normalized by total newspaper article count, then averaged to monthly frequency. The original GPR counts articles matching specific keyword proximity searches. Source: authors' calculations using ProQuest TDM Studio data.

Figure 3: AI-GPR vs. Original GPR Around Key Geopolitical Events



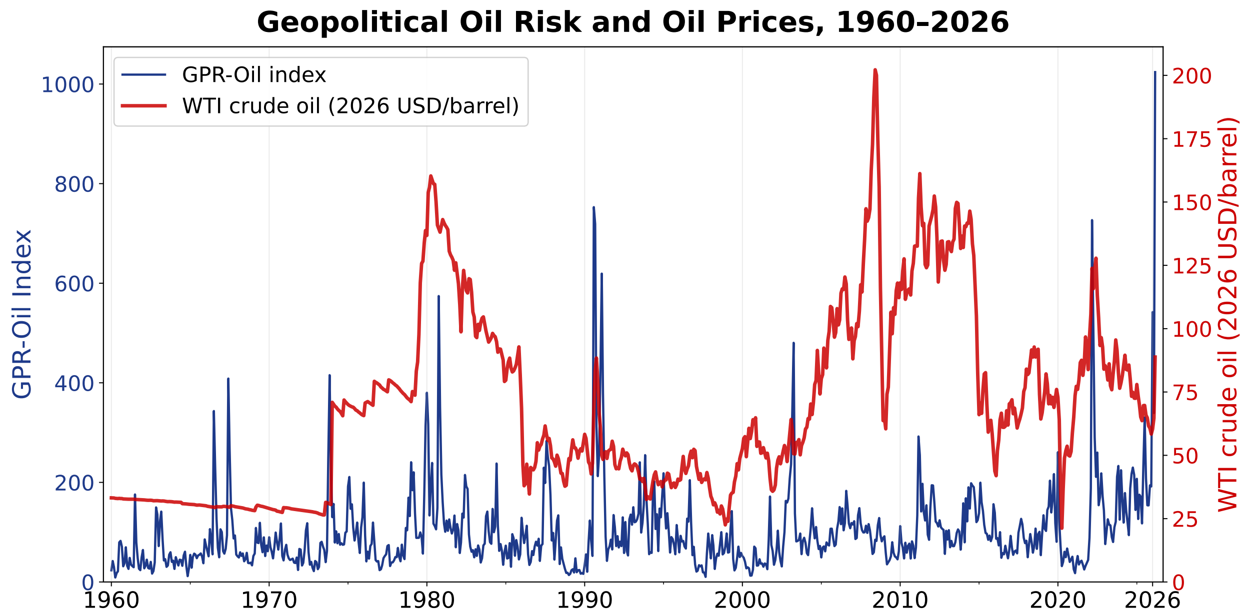
Notes: Each panel shows the daily AI-GPR index (blue, thick) and original GPR index (salmon, thin) around a major geopolitical event. Vertical dashed lines mark the event date. Both indices are at daily frequency (no smoothing) and normalized to a mean of 100 over 1985–2019. Source: authors' calculations.

Figure 4: GEOPOLITICAL THREATS AND ACTS (MONTHLY)



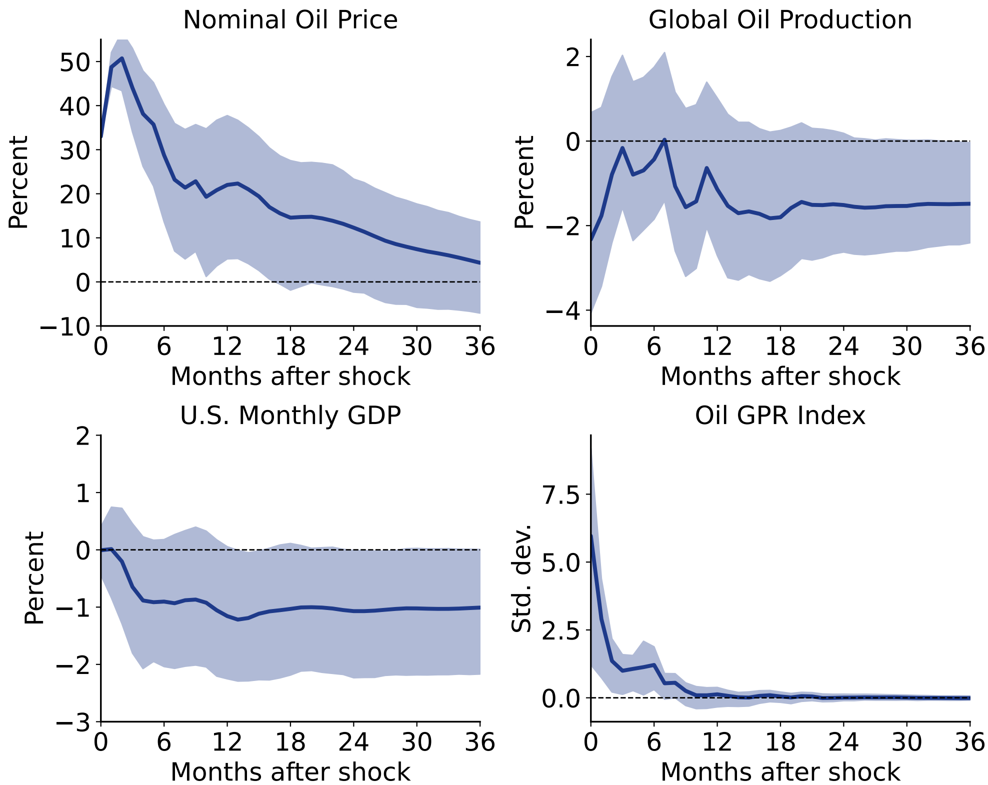
Notes: The threats subindex (orange, thick) and acts subindex (blue, thin) of the AI-GPR index, plotted at monthly frequency, 1960–2025. A second LLM classification layer assigns each geopolitically relevant article (AI-GPR score > 0) to "threat," "act," or "both"; articles labeled "both" contribute to each series. Each series is the 30-day moving sum of the AI-GPR scores of its articles divided by the 30-day moving sum of total newspaper articles, indexed so that each series averages 100 over 1985–2019, then averaged to monthly frequency. Source: authors' calculations using ProQuest TDM Studio data.

Figure 5: GPR-Driven Oil Supply Index and Real Oil Prices



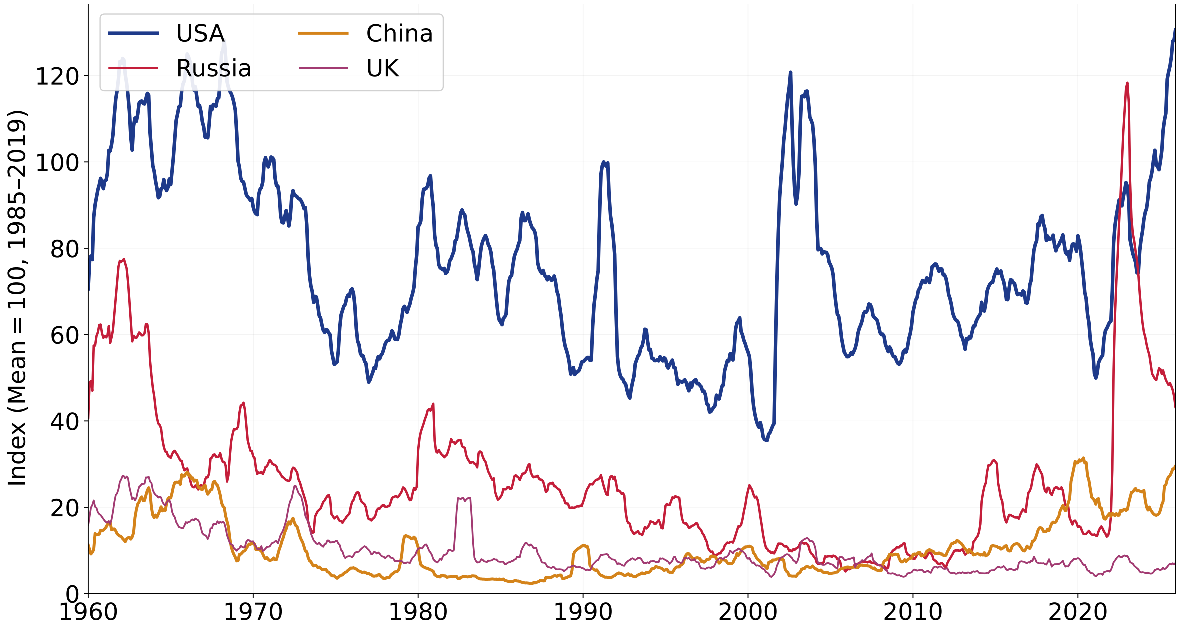
Notes: The dark blue line (left axis) shows the Oil GPR index defined in equation (2), plotted as a 30-day moving average. The red dashed line (right axis) shows the monthly real WTI crude oil price (nominal WTI deflated by the U.S. CPI, expressed in 2019 dollars). Sources: authors' calculations; WTI prices from FRED (series WTISPLC/DCOILWTICO); CPI from FRED (series CPIAUCSL). Last observation is March 14, 2026.

Figure 6: Impulse Responses to a Geopolitical Oil Price Shock



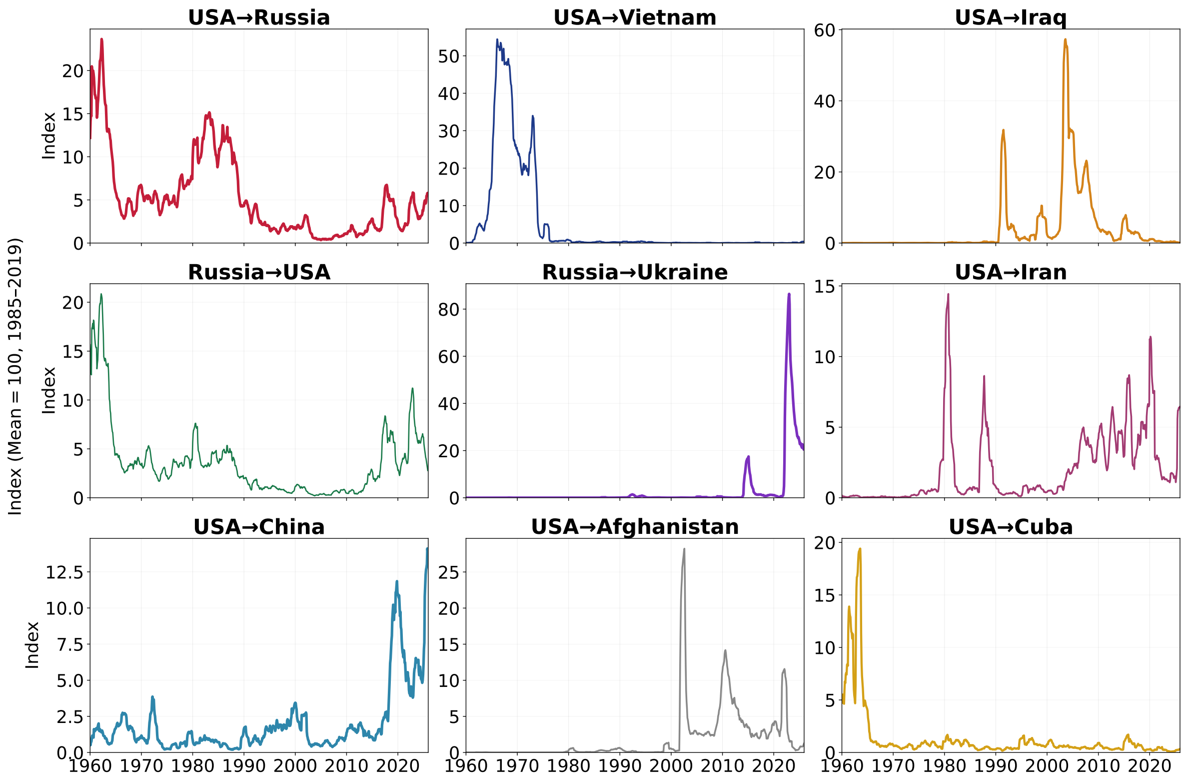
Notes: Impulse responses to a geopolitical oil supply shock identified via a proxy structural VAR. The VAR includes four monthly variables (1975–2024): nominal WTI oil price, global oil production, U.S. monthly GDP (Brave-Butters-Kelley index), and the Oil GPR index, estimated with 12 lags. The external instrument is monthly WTI oil price surprises accumulated on days when the Oil GPR index spikes above two standard deviations. The shock is normalized to generate a 33 percent increase in oil prices on impact, which builds to roughly 50 percent at the peak. Shaded areas represent 80 percent confidence intervals from a wild bootstrap with 500 draws. Sources: WTI prices from FRED; global oil production from the EIA; monthly GDP from FRED (series BBKMGDP); Oil GPR index from authors' calculations.

Figure 7: AI-GPR BY COUNTRY (12-MONTH MOVING AVERAGE)



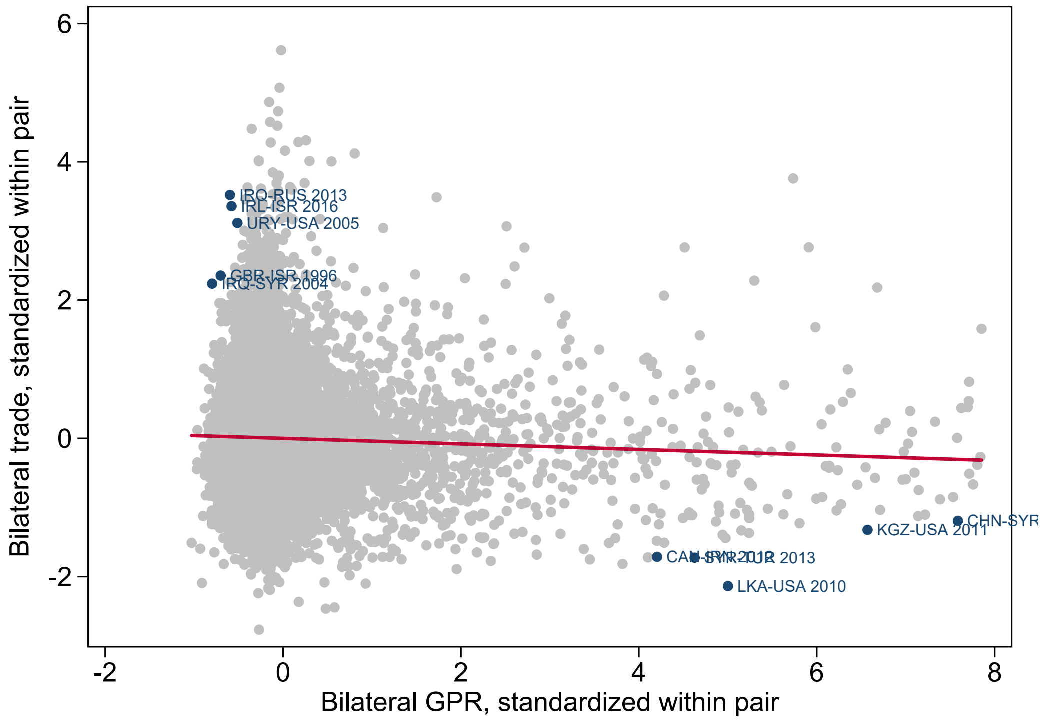
Notes: Monthly AI-GPR contribution for the four most prominent countries across all actor roles (initiator, respondent, spillover), smoothed with a 12-month moving average. Series are scaled to the overall AI-GPR index (mean = 100, 1985–2019). Source: authors' calculations using the second-stage LLM country classifier applied to articles with GPR score > 0.

Figure 8: AI-GPR Bilateral Index: Top Country Pairs (12-Month Moving Average)



Notes: Each panel shows the monthly bilateral AI-GPR sub-index for one directed country pair (initiator → respondent), smoothed with a 12-month moving average. The index aggregates AI-GPR sentiment scores from articles in which the first country is classified as initiator and the second as respondent. Pairs are ranked by average GPR intensity over 1960–2025. Series are scaled to the overall AI-GPR index (mean = 100, 1985–2019). Source: authors' calculations using the second-stage LLM country classifier applied to articles with GPR score > 0.

Figure 9: Bilateral Geopolitical Risk and Trade Flows: Partial Correlation



Notes: Scatter plot of within-pair standardized bilateral trade against within-pair standardized bilateral GPR (one-year lag), both residualized on year fixed effects and log PPP-adjusted GDP of origin and destination. The 100 observations with the largest influence on the estimated slope are labeled with country-pair and year. The line shows the OLS fit. Source: bilateral GPR from this paper; trade from Conte et al. (2023).

A Appendix

A.1 Alternative Model Specifications

We classify a random sample of 1,050 articles using four OpenAI models—GPT-4o-mini, GPT-4o, GPT-5-mini, and GPT-5—together with the open-weight Llama 3.1 8B Instruct, all with the same prompt. We set temperature to zero for the models that support it; the GPT-5 models do not expose an adjustable temperature and run at their default settings. Table A.1 reports the full pairwise correlation matrix.

Table A.1: Cross-Model Score Correlations

	GPT-4o-mini	GPT-4o	GPT-5-mini	GPT-5	Llama
GPT-4o-mini	1.00	0.90	0.87	0.86	0.91
GPT-4o		1.00	0.86	0.87	0.86
GPT-5-mini			1.00	0.94	0.85
GPT-5				1.00	0.85
Llama					1.00

Notes: Pairwise Pearson correlations of geopolitical risk scores assigned by each model to a random sample of 1,050 newspaper articles. All models use the same classification prompt, at temperature zero where supported; the GPT-5 models run at their default settings. “Llama” denotes the open-weight Llama 3.1 8B Instruct model.

All pairwise correlations exceed 0.85, with the highest agreement (0.94) between GPT-5-mini and GPT-5. The mean score ranges from 0.10 (GPT-4o) to 0.17 (GPT-5-mini), reflecting modest differences in calibration that wash out after normalization. These results confirm that the AI-GPR index is robust to model choice—across two generations of OpenAI models, across model sizes, and across proprietary and open-weight model families.

A.2 Details of the Human Audit

This appendix gives the full results of the human validation summarized in Section 2.6. We independently scored a random sample of 1,050 articles—drawn from the *New York Times* and the *Washington Post* over January 1960 to December 2025—using the same rubric supplied to the model, which assigns each article a geopolitical risk score between 0 and 1. We then compared our scores with those produced by GPT-4o-mini on the first 2,000 characters of each article.

Agreement between the human and machine scores is high. The Pearson correlation between the two is 0.88, and the Spearman rank correlation is 0.87. The scores coincide exactly for 76 percent of articles and lie within one step of the scoring scale (0.2) for 98 percent. The mean absolute difference is 0.05, small relative to a human-score standard deviation of 0.19. Of the 247 articles on which the scores differ, 93 percent differ by exactly one step—typically a defensible judgment call at a rubric boundary, such as whether an ongoing regional conflict constitutes “significant tensions” (0.4) or “substantial discussion of major war escalation risks” (0.6), rather than a misreading of the article. Table A.2 reports the full cross-tabulation. The machine scores are slightly higher on average than ours (0.13 versus 0.11): the model scores above the human in 181 cases and below in 66. This gap is stable across decades, from about 0.01 in the 1960s–1980s to about 0.03 from the 1990s onward, so the level difference is unlikely to distort the time variation of the index.

Table A.2: Human vs. AI Score Cross-Tabulation

	AI score				
Human score	0.0	0.2	0.4	0.6	0.8
0.0	692	31	7	0	0
0.2	38	32	65	5	0
0.4	4	8	38	66	0
0.6	1	0	8	40	6
0.8	0	0	1	5	2
1.0	0	0	0	0	1

Notes: Cross-tabulation of human and GPT-4o-mini scores for the 1,050 audited articles. Rows are human scores, columns are model scores. The model never assigned a score above 0.8 (one article scored 0.3 by the model is grouped with 0.4). Off-diagonal mass is concentrated on the first off-diagonal, that is, one-step disagreements.

Reading the disagreeing articles and recording notes on each case, we group the discrepancies into five recurring patterns. *First*, the model treats economic and financial tensions between countries as geopolitical risk. The largest source of upward bias is cross-border economic conflict: articles on a proposed U.S. tariff on Japanese goods (1971), the Latvian currency crisis (2009), Iceland's refusal to repay British and Dutch lenders (2010), and strained Taiwan–China business ties (2000) were scored 0.4 to 0.6 by the model and 0 to 0.2 by us. The rubric defines geopolitical risk around war, terrorism, and military tension; trade and financial disputes between friendly nations strain "ties" in a literal sense but carry essentially no risk of armed conflict, and a human coder discounts them accordingly. Roughly 50 of the 181 cases in which the model scores higher are of this type, consistent with the model anchoring on surface vocabulary ("threat," "retaliation," "frayed ties") rather than the military nature of the underlying risk.

Second, the model treats domestic unrest and domestic security as geopolitical risk. Purely internal events with security overtones—the 1991 Crown Heights riots, labor strife framed around a general strike, an article on the disciplining of unruly airline passengers before the 2021 inauguration—were scored 0.4 to 0.6 by the model. The prompt explicitly instructs that domestic political events should score zero absent international implications, but the model appears to weight words such as "violence," "riot," and "security" heavily even in domestic settings.

Third, the model makes temporal-attribution errors in both directions. The rubric instructs that retrospective coverage of old events should score zero unless current risks are mentioned. In some cases the model fails to recognize aftermath coverage as backward-looking: a 2004 article on legal turmoil in a Detroit terrorism prosecution (events eight months past) received 0.6, and a 2002 piece on routine airport security received 0.4. In other cases it applies the retrospective rule too aggressively: human-interest profiles published days after the 1993 World Trade Center bombing and the 1998 embassy bombings were scored 0 by the model even though the attacks were current events signaling elevated terrorism risk. The model evaluates the 2,000-character snippet in isolation, whereas a human coder knows the article's date and the surrounding history.

Fourth, the human coder sometimes credits context the snippet only implies. A 1961 wire item on a Navy missile contract was scored 0.2 by us (arms development at the height of the Cold War) and 0 by the model, which is arguably the more faithful reading of the prompt's instruction to classify based only on what an article states or strongly implies. A brief 2003 item on the capture of Ali Hassan al-Majid scored 0.6 in our coding—it presupposes the ongoing Iraq war—and 0 from the model. Reports on the humanitarian consequences of conflict, such as a 1973 piece on war-induced famine in Cambodia, scored higher in our coding because we treated humanitarian devastation and the implied risk of escalation as risk content while the model adhered to the narrower definition. In these cases the human is more generous than the prompt strictly licenses, so the disagreement reflects the rubric as much as the model.

Fifth, the model occasionally scores the backdrop rather than the subject of an article. When a major conflict serves as background for a story about something else—a 2023 piece on how the war in Ukraine affects livelihoods in a Norwegian border town, a 2017 color piece on the mood in a Beirut neighborhood—the model scores the conflict (0.6) while we score the article (0.2). The conflicts are real, but the articles convey little incremental information about war risk; a human coder weights the subject of the article, the model the most alarming entity mentioned in it.

Finally, the model compresses the top of the scale. It assigned 0.8 to only three articles and never scored above 0.8, whereas we used 0.8 or 1.0 nine times; of those nine, the model scored six at 0.6 or below. Conditional on an article being high-risk by human assessment (0.6 or above), the model's mean score is 0.59 against a human mean of 0.63. The compression is mild and affects few observations, but it implies that the machine index understates the relative intensity of the most extreme episodes. Taken together, and given a correlation of 0.88 with one-step agreement of 98 percent, the audit supports treating the LLM scores as a reliable proxy for careful human coding, with the caveats that the index runs slightly hot at the bottom of the scale and slightly cold at the top.

A.3 Keyword-Based GPR Index

As described in Section 3, we construct a keyword-based GPR index using the Caldara and Iacoviello (2022) methodology on the same three newspapers and time period as the AI-GPR. The search terms are organized into eight categories, each combining a geopolitical term group with a risk/threat term group using the N/2 proximity operator (words must appear within 2 words of each other). The full search string is:

```
GPR ORIGINAL SEARCH QUERY:
(war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR
revolt OR coup OR geopolitical) N/2 (risk* OR warn* OR fear* OR danger* OR threat*
OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR imminen* OR
inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (peace OR truce OR armistice OR treaty OR parley) N/2 (menace* OR reject* OR
threat* OR peril* OR boycott* OR disrupt*)
OR (military OR troops OR missile* OR ``arms'' OR weapon* OR bomb* OR warhead*)
AND (buildup* OR build-up* OR blockad* OR sanction* OR embargo OR quarantine OR
ultimatum OR mobiliz*)
OR (``nuclear war'' OR ``atomic war'' OR ``nuclear missile*'' OR ``nuclear bomb*''
OR ``atomic bomb*'' OR ``h-bomb*'' OR ``hydrogen bomb*'' OR ``nuclear weapon*'')
AND (risk* OR warn* OR fear* OR danger* OR threat* OR doubt* OR crisis OR troubl*
OR disput* OR concern* OR tension* OR imminen* OR inevitable OR footing OR menace*
OR brink OR scare OR peril*)
```

```
OR (terroris* OR guerrilla* OR hostage*) N/2 (risk* OR warn* OR fear* OR danger*
OR threat* OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR
imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR
revolt OR coup OR geopolitical) N/2 (begin* OR begun OR began OR outbreak OR
‘‘broke out’’ OR breakout OR start* OR declar* OR proclamation OR launch*)
OR (allie* OR enem* OR foe* OR army OR navy OR aerial OR troops OR rebels OR
insurgen*) N/2 (drive* OR shell* OR advance* OR offensive OR invasion OR invad* OR
clash* OR attack* OR raid* OR launch* OR strike*)
OR (terroris* OR guerrilla* OR hostage*) N/2 (act OR attack OR bomb* OR kill* OR
strike* OR hijack*)
NOT (movie* OR film* OR museum* OR anniversar* OR obituar* OR memorial* OR arts OR
book OR books OR memoir* OR ‘‘price war’’ OR game OR story OR history OR veteran*
OR tribute* OR sport OR music OR racing OR cancer OR ‘‘real estate’’ OR mafia OR
trial OR tax)
```

The resulting index is the ratio of articles matching the proximity search to total articles published on each date, normalized to have a mean of 100 over 1985–2019. This is the “GPR (Original)” series reported throughout the paper.

A.4 GPR Subindexes

We use the following prompts to classify articles for the GPR subindexes.

A.4.1 Threats and Acts GPR

The threats and acts classification prompt identifies the nature of geopolitical risk. Relevant articles are labeled based on whether they predominantly discuss anticipated risk (threats), realized risk (acts), or both.

```
THREATS AND ACTS CLASSIFICATION PROMPT:
Analyze this newspaper article about geopolitical risk.  This article has been
identified as discussing an adverse geopolitical event (Geopolitical risk).
Geopolitical risk (GPR) encompasses adverse geopolitical events and associated
risks, including:
- Wars (nuclear and conventional)
- Terrorist threats and acts
- Military buildups and tensions
- International crises and conflicts
TASK: Classify the nature of geopolitical risk discussed in this article.
The three types of geopolitical risk nature are defined as follows:
  1. threat:  Risks, tensions, warnings, or potential escalations related to
     adverse geopolitical events
  2. act:  Actual adverse geopolitical events that have occurred or are actively
     occurring (wars, attacks, terrorist acts, invasions, coups, etc.)
```

```
3. both:  Substantive discussion of both realized events AND associated
   threats/risks

Rules:
- Choose ONLY ONE: 'threat', 'act', or 'both'
- Be precise:  only choose 'both' if the article substantially discusses BOTH
  temporal dimensions
- Brief mentions of past context while focusing on future risks = 'threat'
- Brief mentions of future risks while focusing on current/past events = 'act'
```

A.4.2 Oil GPR

The Oil GPR classification prompt first identifies whether an article talks about potential or realized oil supply disruptions. Relevant articles are then assigned regional labels based on where the oil supply disruption/threat is and which regions are affected. Multiple countries can be listed under the regional label.

```
OIL-GPR CLASSIFICATION PROMPT:
Analyze this newspaper article about geopolitical risk.
TASK 1:  Determine if this article discusses events that could disrupt oil or
energy supplies.  Oil/Energy disruption includes:
- Military conflicts in oil-producing regions
- Sanctions on oil-producing countries
- Attacks on oil infrastructure, pipelines, refineries, or shipping routes
- Political instability in major oil producers
- Blockades of oil shipping lanes (Strait of Hormuz, Suez Canal, etc.)
- Natural disasters affecting oil production
- Oil embargoes or export restrictions
- Terrorist attacks on energy infrastructure
Answer "yes" if the article discusses events that could affect oil/energy
production, transportation, or supply.  Answer "no" if the article discusses
geopolitical events unlikely to affect oil/energy markets.

TASK 2:  If "yes", identify which oil/energy producing region(s) are affected.
Select one or more from this list ONLY: Middle_East, Russia, USA, Venezuela,
North_Africa, West_Africa, Central_Asia, North_Sea, Canada, Mexico, Latin_America,
Southeast_Asia, China, Other
REGIONAL MAPPING RULES:
- Use 'Middle_East' for:  Saudi Arabia, Iran, Iraq, Kuwait, UAE, Qatar, Bahrain,
  Oman, Yemen, Israel, Syria, Lebanon
- Use 'Russia' for:  Russia, Soviet Union, USSR, Russian Federation
- Use 'USA' for:  United States, America, USA, U.S. (as oil producer, not just
  consumer)
- Use 'Venezuela' for:  Venezuela
- Use 'North_Africa' for:  Libya, Algeria, Egypt, Tunisia
```

```
- Use 'West_Africa' for:  Nigeria, Angola, Equatorial Guinea, Gabon, Congo
- Use 'Central_Asia' for:  Kazakhstan, Azerbaijan, Turkmenistan, Uzbekistan
- Use 'North_Sea' for:  Norway, UK (when referring to North Sea oil production)
- Use 'Canada' for:  Canada (oil sands, Alberta, etc.)
- Use 'Mexico' for:  Mexico
- Use 'Latin_America' for:  Other Latin American oil producers (Brazil, Colombia,
  Ecuador, Argentina)
- Use 'Southeast_Asia' for:  Indonesia, Malaysia, Brunei, Vietnam
- Use 'China' for:  China (as oil producer)
- Use 'Other' for:  Any other oil/energy producing region not listed above
IMPORTANT:
- Focus on regions WHERE the disruption is occurring, not where it's being
  discussed
- Include ALL regions mentioned that could experience oil/energy disruption
- If article mentions global impact or multiple regions, list all relevant ones
- If uncertain whether it affects oil/energy, err on the side of ``yes" if the
  region is a major producer
```

A.4.3 Country GPR (and Event)

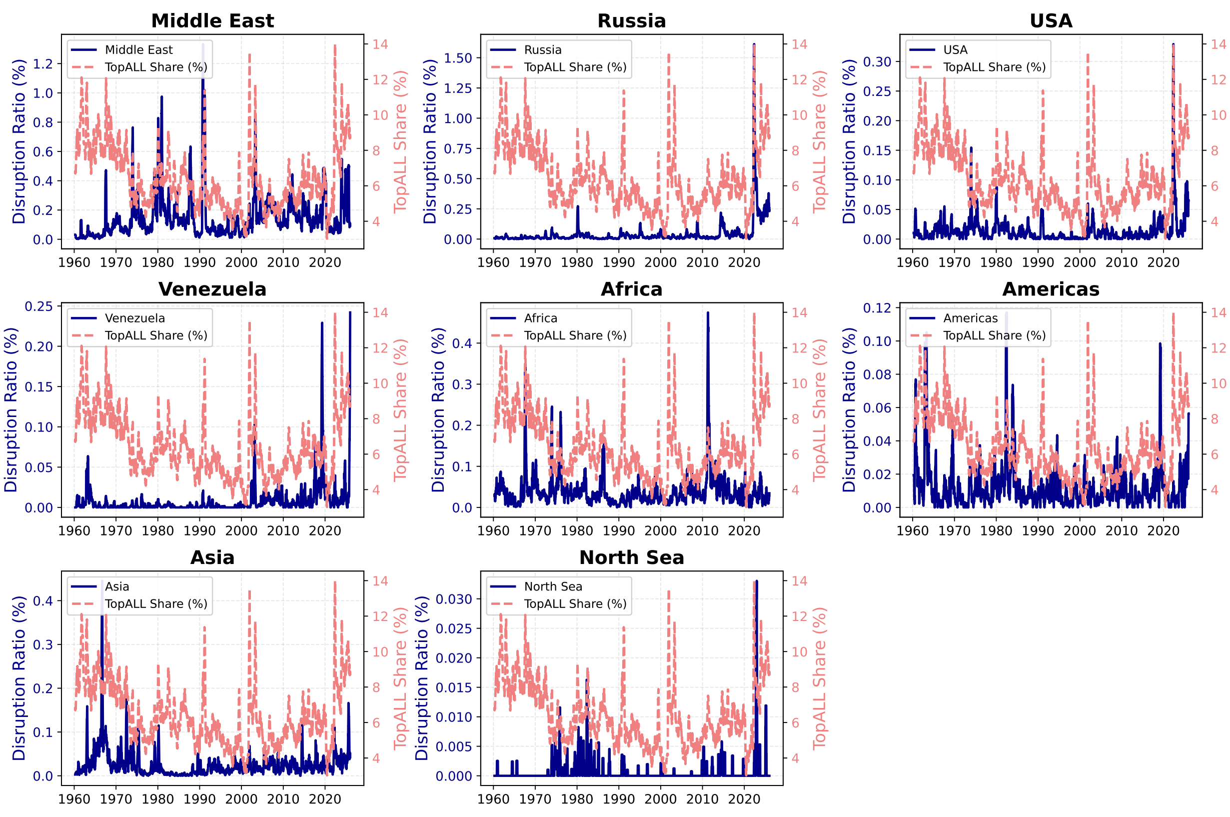
The country classifier prompt identifies the main countries involved in a geopolitical incident as portrayed in a given article. Relevant countries are assigned one of three actor labels indicating whether they are an initiator, a respondent, or a spillover party. The prompt also assigns an event-type label for the article's central geopolitical conflict.

```
COUNTRY GPR:
Analyze this newspaper article about a geopolitical risk event.
TASK: Identify the key country-level actors in this geopolitical event and assign
each a role.  Roles are defined as follows:
- initiator:  the country that launched or triggered the hostile action (military
  attack, invasion, sanctions, blockade, etc.)
- respondent:  the country that is the direct object of the hostile action
- spillover:  a country not directly involved but significantly affected (energy
  shock, refugee flows, trade disruption, etc.)
Rules:
- List up to 5 actors total; include only countries with a meaningful role in THIS
  article
- Use standard country names (e.g.  "United States", "Russia", "Saudi Arabia",
  "Israel")
- If a supra-national body (NATO, UN, EU) is the dominant actor, use its name
- Assign exactly one role per actor
- When initiation is genuinely contested or unclear (civil wars, mutual
  escalations, disputed territories), assign both main parties the role of
  initiator
```

```
- If the article is too vague to identify actors, return an empty actors list
Respond in JSON format only:
{"actors": [
  {"country": "<name>", "role": "<role>"},
  ...
],
"event_type": "<one of: military_conflict, sanctions, terrorism,
diplomatic_tension, coup, civil_war, nuclear_threat, other>"}
```

A.5 GPR-Driven Oil Disruptions by Region

Figure A.1: GPR-Driven Oil Disruptions by Region



Notes: Each panel shows the regional Oil GPR index (90-day moving average) for a specific geographic region. The index is computed as the ratio of sentiment-weighted articles about oil disruptions in that region to total newspaper articles. The light red dashed line in each panel shows the overall AI-GPR share for reference. Regions are: Middle East, Russia, USA, Venezuela, Africa (North + West), Americas (Canada, Mexico, Latin America), Asia (Central, Southeast, China), and North Sea. Source: authors' calculations using the two-stage LLM classification described in Section 4.2.